HW3

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Contents

Problem 1

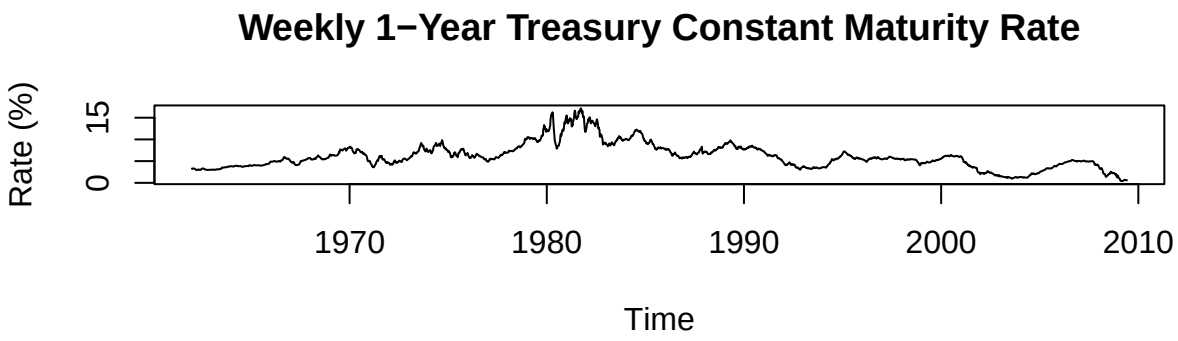
a)

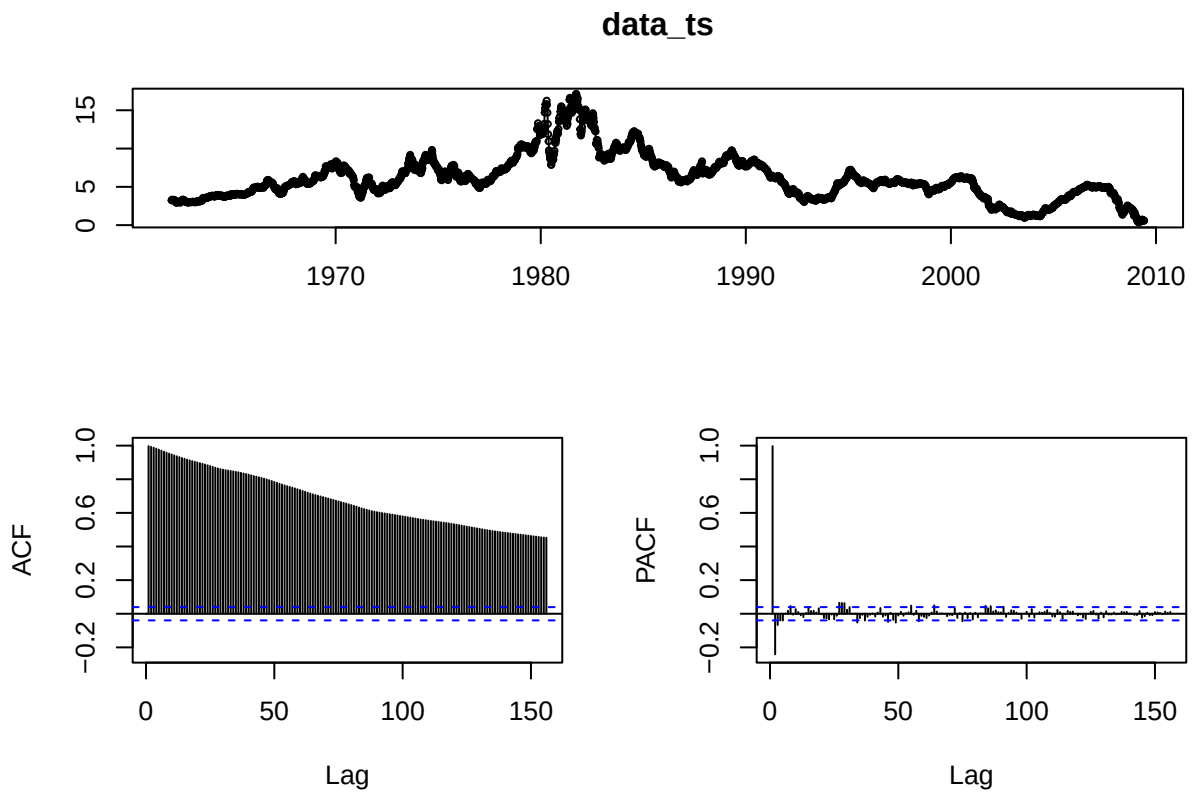
```
data=read.table(file = "../../data/w-gs1yr-2.txt", header = TRUE)
head(data)
```

```
##   year mon day rate
## 1 1962   1   5 3.24
## 2 1962   1  12 3.32
## 3 1962   1  19 3.29
## 4 1962   1  26 3.26
## 5 1962   2   2 3.29
## 6 1962   2   9 3.29
```

```
data_ts = ts(data$rate, start = c(1962, 1), frequency = 52)

par(mfrow = c(2, 1))
plot(data_ts, main = "Weekly 1-Year Treasury Constant Maturity Rate", xlab = "Time", ylab = "Rate (%)")
tsdisplay(data_ts)
```


Weekly 1−Year Treasury Constant Maturity Rate
Rate (%)
0
15
1970
1980
1990
2000
2010
Time



The ACF starts very high and then decays slowly. The PACF cuts off at lag 2 or 3 which would indicate that this is an AR(2) or AR(3). b)

```
ar2=arima(data_ts, order = c(2, 0, 0))
ar4=arima(data_ts, order = c(4, 0, 0))
ar3=arima(data_ts, order = c(3, 0, 0))
summary(ar2)
```

```
##
## Call:
## arima(x = data_ts, order = c(2, 0, 0))
##
## Coefficients:
##          ar1      ar2  intercept
##       1.3434  -0.3458     6.0621
## s.e.  0.0189   0.0189     1.3612
##
## sigma^2 estimated as 0.03151:  log likelihood = 761.22,  aic = -1514.44
##
## Training set error measures:
##                        ME      RMSE       MAE        MPE     MAPE      MASE
## Training set -0.0006784007 0.1775171 0.1050281 -0.1586663 1.832503 0.9398335
##                      ACF1
## Training set -0.02001622
```

```r
summary(ar3)
```

```
##
## Call:
## arima(x = data_ts, order = c(3, 0, 0))
##
## Coefficients:
##          ar1      ar2      ar3  intercept
##       1.3232  -0.2676  -0.0582     6.0939
## s.e.  0.0201   0.0329   0.0201     1.2843
##
## sigma^2 estimated as 0.03141:  log likelihood = 765.39,  aic = -1520.78
##
## Training set error measures:
##                         ME     RMSE       MAE        MPE     MAPE      MASE
## Training set -0.0007161213 0.177217 0.1050358 -0.1601303 1.835013 0.9399022
##                      ACF1
## Training set -0.002522984
```

```r
summary(ar4)
```

```
##
## Call:
## arima(x = data_ts, order = c(4, 0, 0))
##
## Coefficients:
##          ar1      ar2     ar3      ar4  intercept
##       1.3208  -0.2798  0.0011  -0.0449     6.0496
## s.e.  0.0201   0.0333  0.0333   0.0201     1.2052
##
## sigma^2 estimated as 0.03134:  log likelihood = 767.89,  aic = -1523.79
##
## Training set error measures:
##                         ME      RMSE       MAE        MPE     MAPE      MASE
## Training set -0.0005626719 0.1770376 0.1047909 -0.1580621 1.827135 0.9377107
##                      ACF1
## Training set -0.00235302
```

```r
AIC(ar2, ar3, ar4)
```

```
##     df       AIC
## ar2  4 -1514.436
## ar3  5 -1520.784
## ar4  6 -1523.788
```

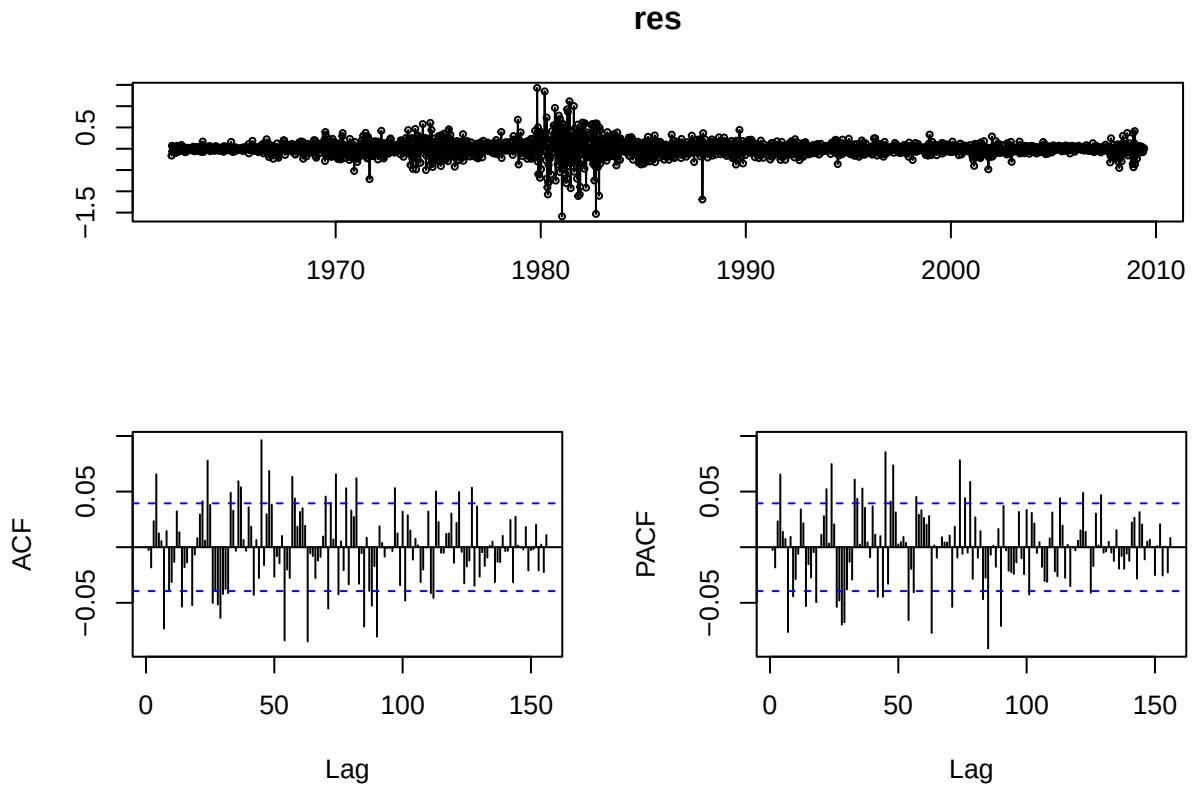
```r
BIC(ar2, ar3, ar4)
```

```
##     df       BIC
## ar2  4 -1491.193
## ar3  5 -1491.730
## ar4  6 -1488.924
```

After comparing the AIC and BIC, the AR(4) has the lowest AIC while the AR(3) has the lowest BIC. I prefer the AR(3) because BIC penalizes extra parameters more heavily and the improvement in AIC from adding a fourth lag is small.

c)

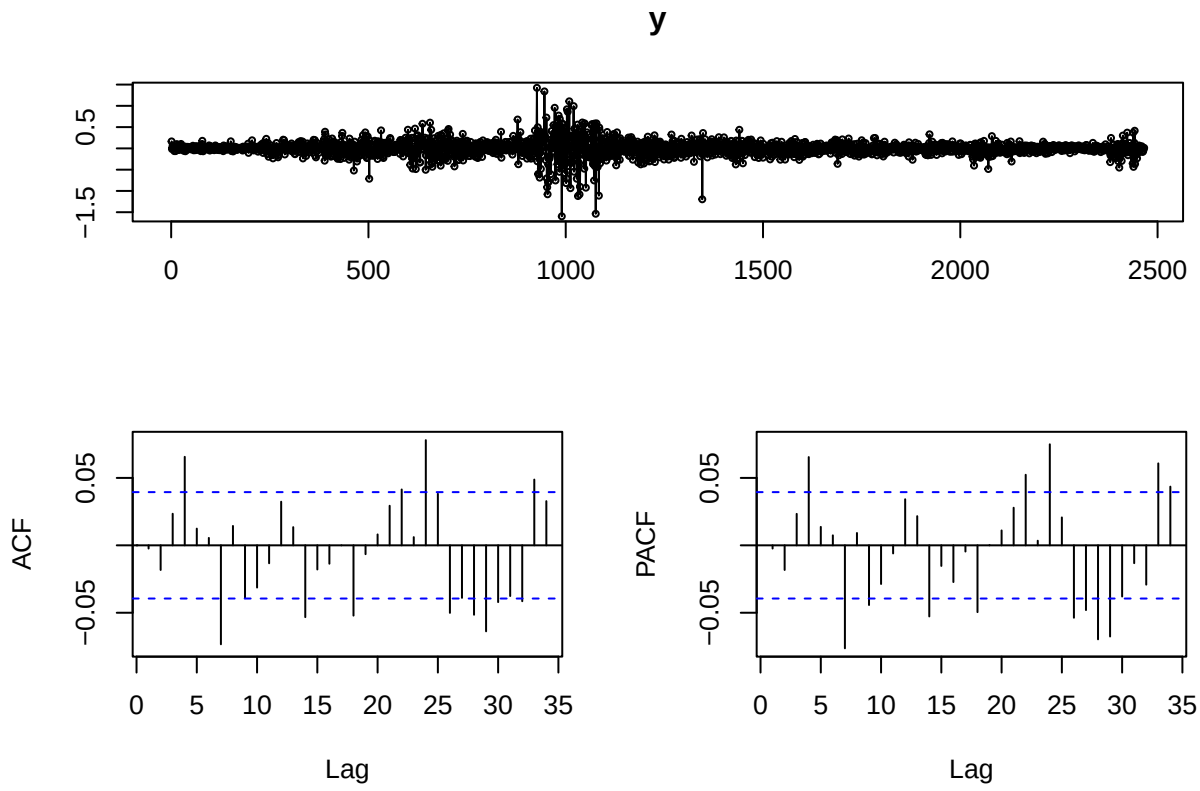
```
res=residuals(ar3)
tsdisplay(res)
```



The residuals seem to be stationary, however, the ACF and PACF both have several significant lags. This indicates that there are dynamics in play that are not captured by the model.

d)

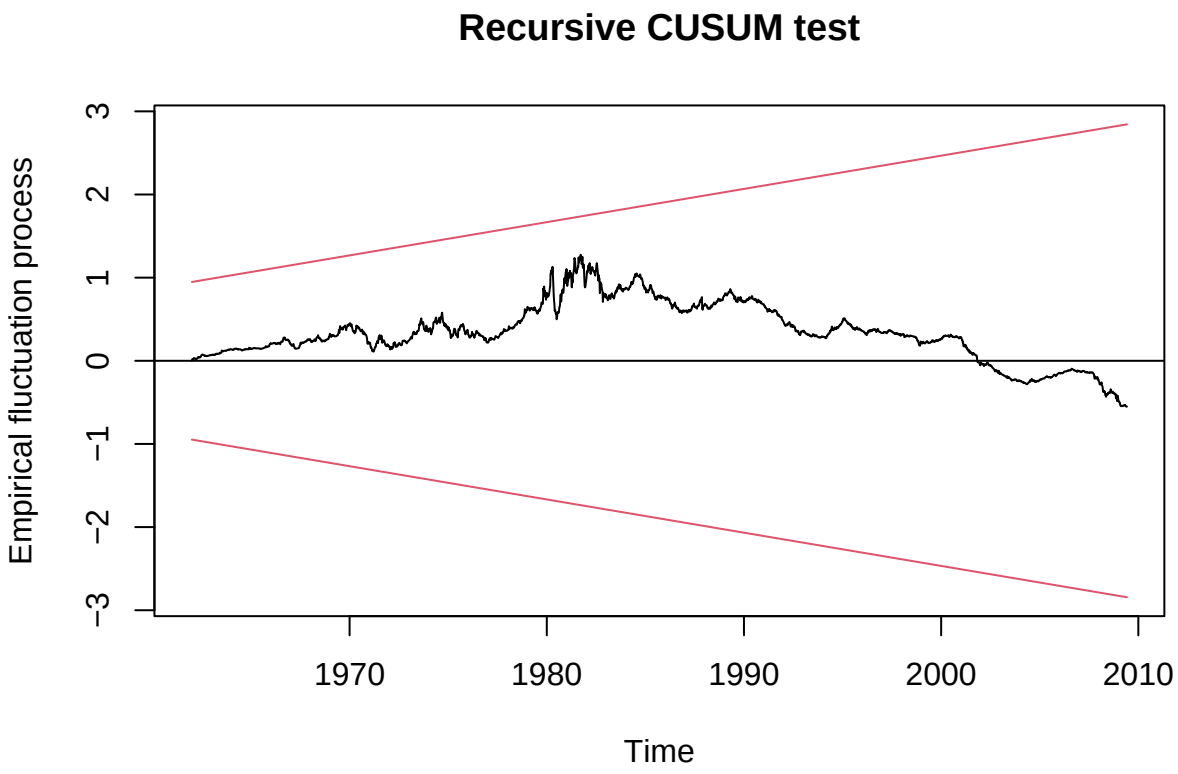
```
y=recresid(ar3$res~1)
tsdisplay(y)
```



There are many spikes in the ACF and PACF and they also show signs of some sort of cycles as they seem to oscillate. This indicates further dynamics that need to be picked up.

e)

```
plot(efp(ar3$res~1, type = "Rec-CUSUM"))
```



The Cumulative Sum stays within the lines which means that there is not a structural breakdown evident.

f)

```
rmodel=auto.arima(data_ts)
summary(rmodel)
```

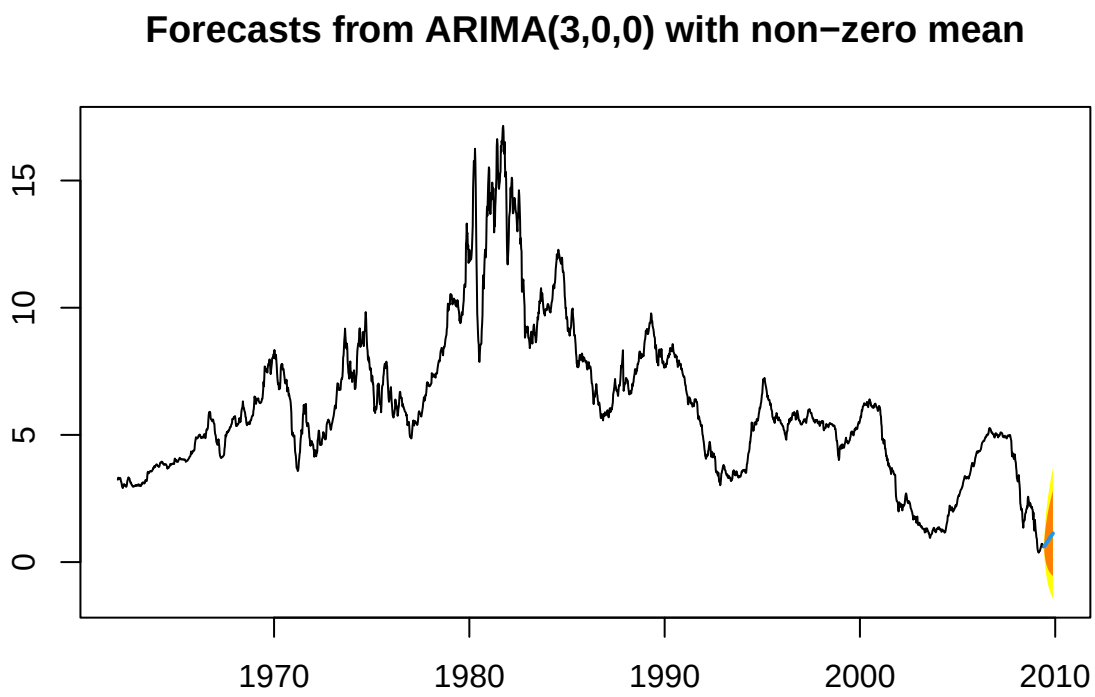
```
## Series: data_ts
## ARIMA(1,1,2)
##
## Coefficients:
##          ar1      ma1      ma2
##       0.6284  -0.3065  -0.0527
## s.e.  0.0642   0.0675   0.0299
##
## sigma^2 = 0.03143:  log likelihood = 768.32
## AIC=-1528.65   AICc=-1528.63   BIC=-1505.41
##
## Training set error measures:
##                        ME      RMSE       MAE         MPE     MAPE       MASE
## Training set -0.0006210918 0.1771539 0.1046884 -0.05084758 1.820023 0.07539674
##                       ACF1
## Training set -0.0002601686
```

After running auto.arima, it chose an arima(1,1,2) which means that there is an AR(1), MA(2), and the data was differenced once. My model was an AR(3), and did not perform as well as the auto.arima function

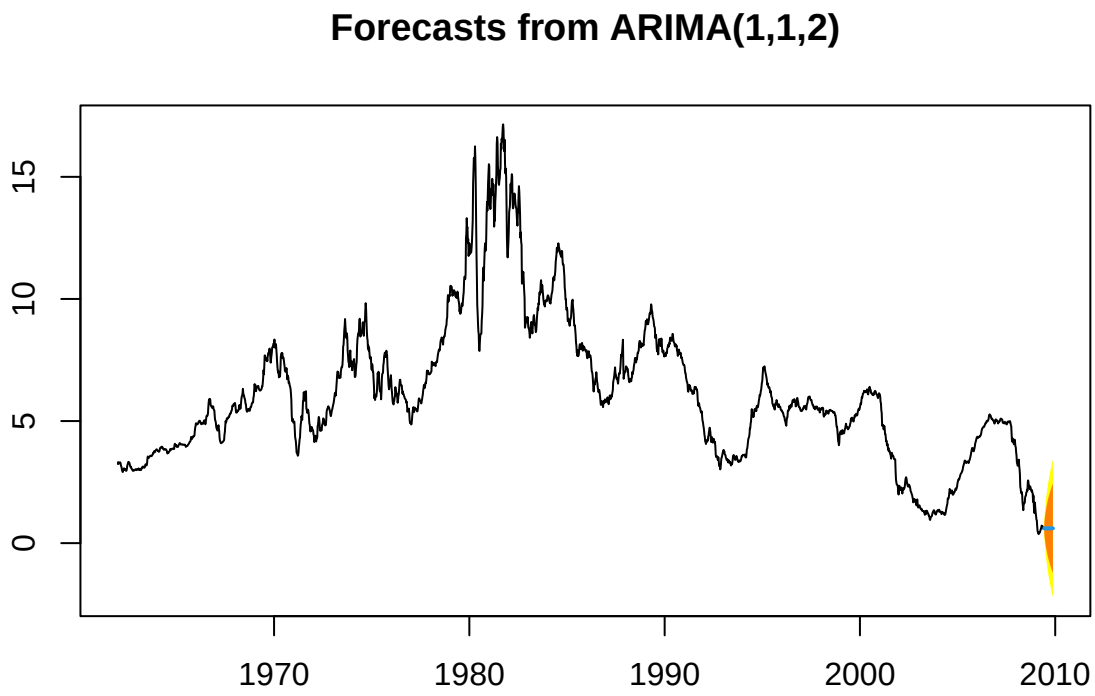
according to AIC and BIC, although these criteria are not strictly comparable between a model in levels and a model in first differences.

g)

```
plot(forecast(ar3,h=24),shadecols="oldstyle")
```




```
plot(forecast(rmodel,h=24),shadecols="oldstyle")
```

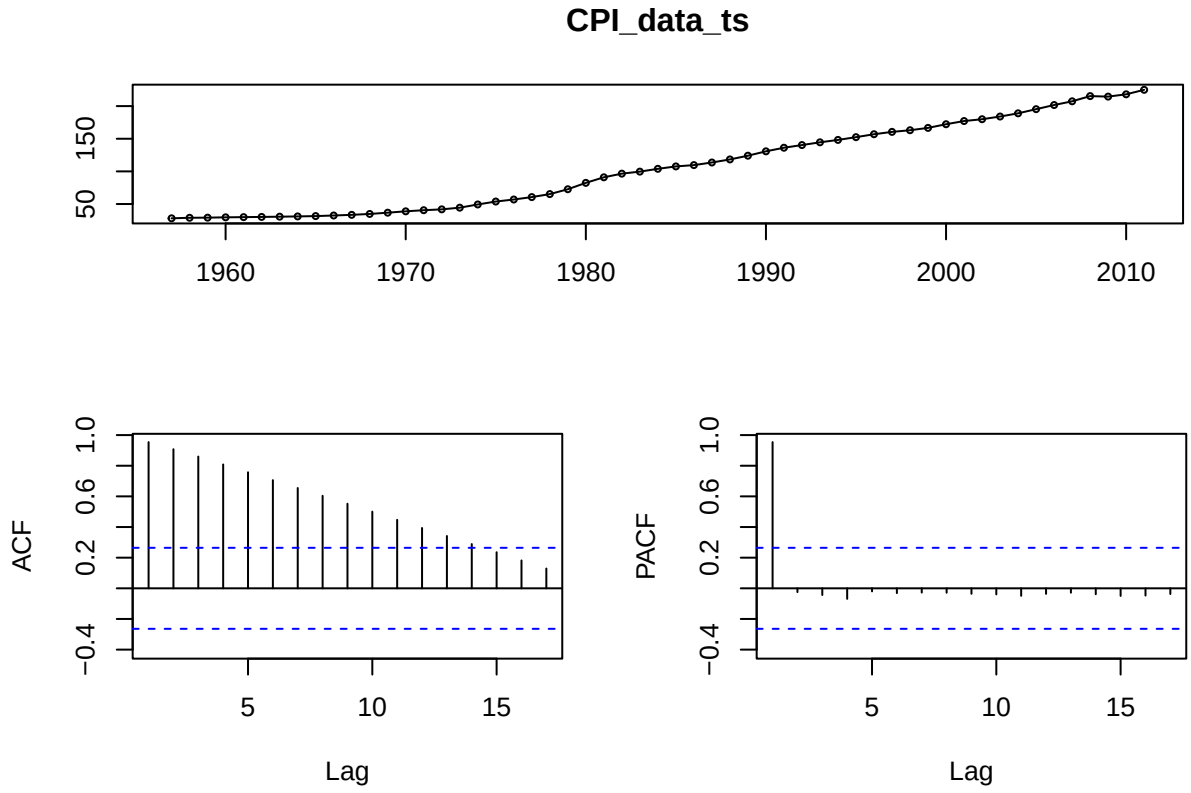


The forecast of the ar(3) shows a more positive upward trend compared to the arima(1,1,2). Besides that, both are quite similar except for more movement in the ar3.

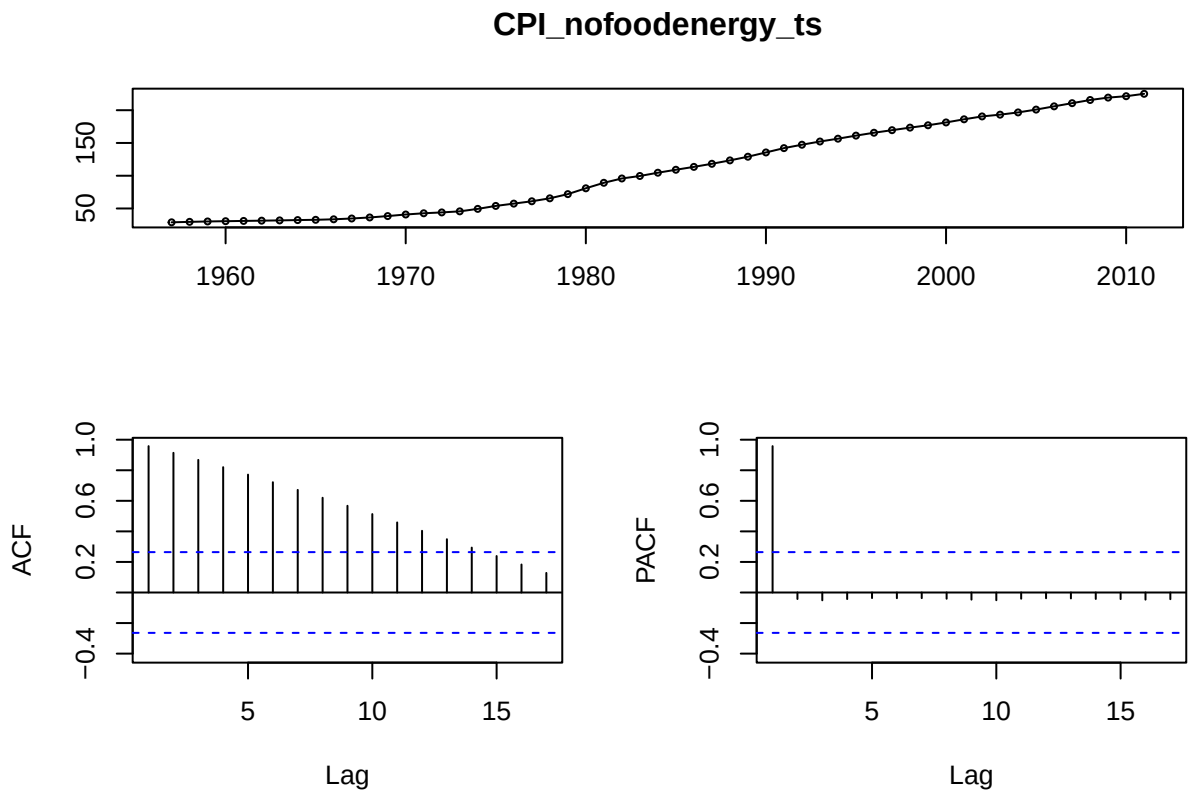
Problem 2

```
CPI_data=read_excel("../../data/Chapter7_Exercises_Data.xls", sheet="Exercise 8")
CPI_data_ts=ts(CPI_data[2], start = c(1957, 1), frequency = 1)
CPI_nofoodenergy_ts=ts(CPI_data[4], start = c(1957, 1), frequency = 1)

tsdisplay(CPI_data_ts)
```




```
tsdisplay(CPI_nofoodenergy_ts)
```




```
cpi_model=auto.arima(CPI_data_ts)
cpi_nofoodenergy_model=auto.arima(CPI_nofoodenergy_ts)
summary(cpi_model)
```

```
## Series: CPI_data_ts
## ARIMA(0,2,2)
##
## Coefficients:
##           ma1      ma2
##       -0.2919  -0.3680
## s.e.   0.1363   0.1396
##
## sigma^2 = 2.998:  log likelihood = -103.54
## AIC=213.08   AICc=213.57   BIC=218.99
##
## Training set error measures:
##                      ME     RMSE      MAE       MPE    MAPE      MASE
## Training set 0.2269873 1.667221 1.050281 0.4240755 1.09854 0.2859047
##                      ACF1
## Training set -0.03114694
```

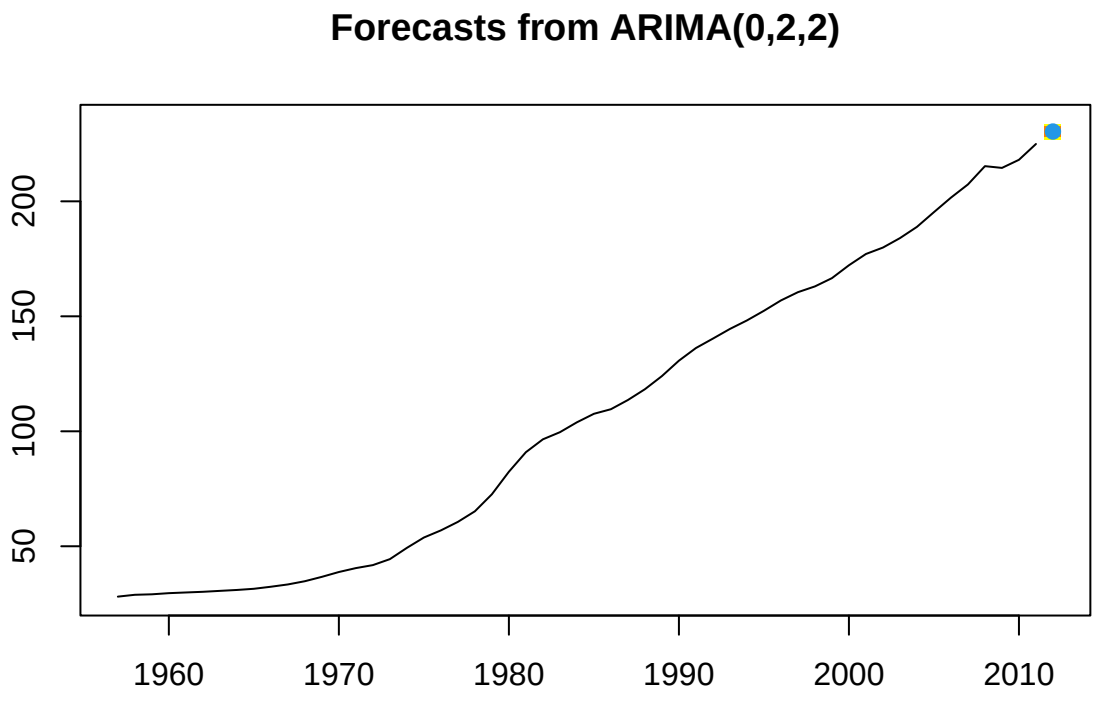
```
summary(cpi_nofoodenergy_model)
```

```
## Series: CPI_nofoodenergy_ts
```

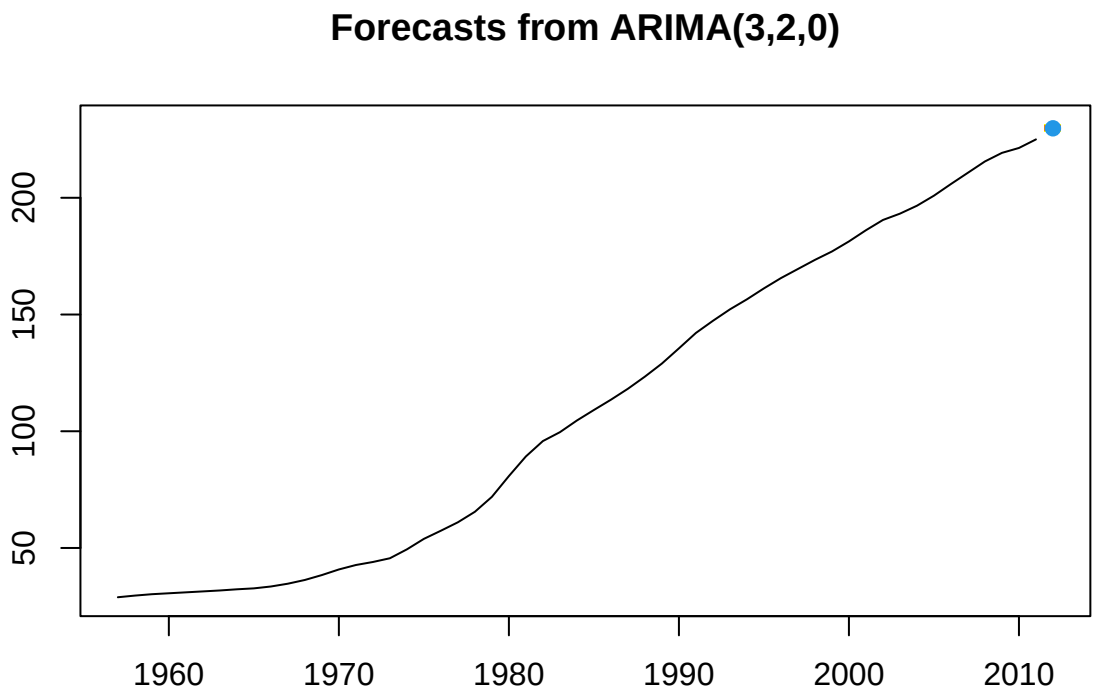
```
## ARIMA(3,2,0)
##
## Coefficients:
##          ar1      ar2      ar3
##       0.2360  -0.2495  -0.2657
## s.e.  0.1365   0.1393   0.1376
##
## sigma^2 = 0.7541:  log likelihood = -66.44
## AIC=140.88   AICc=141.71   BIC=148.76
##
## Training set error measures:
##                       ME      RMSE       MAE      MPE      MAPE      MASE
## Training set 0.08168398 0.8279549 0.6084243 0.203859 0.6874307 0.1675348
##                       ACF1
## Training set -0.002435562
```

It looks as if all CPI is more difficult to predict because there are slightly more dynamics in the ACF and PACF compared to the CPI without food and energy. The summary statistics such as RMSE agree with this as RMSE is lower for the CPI without food and energy.

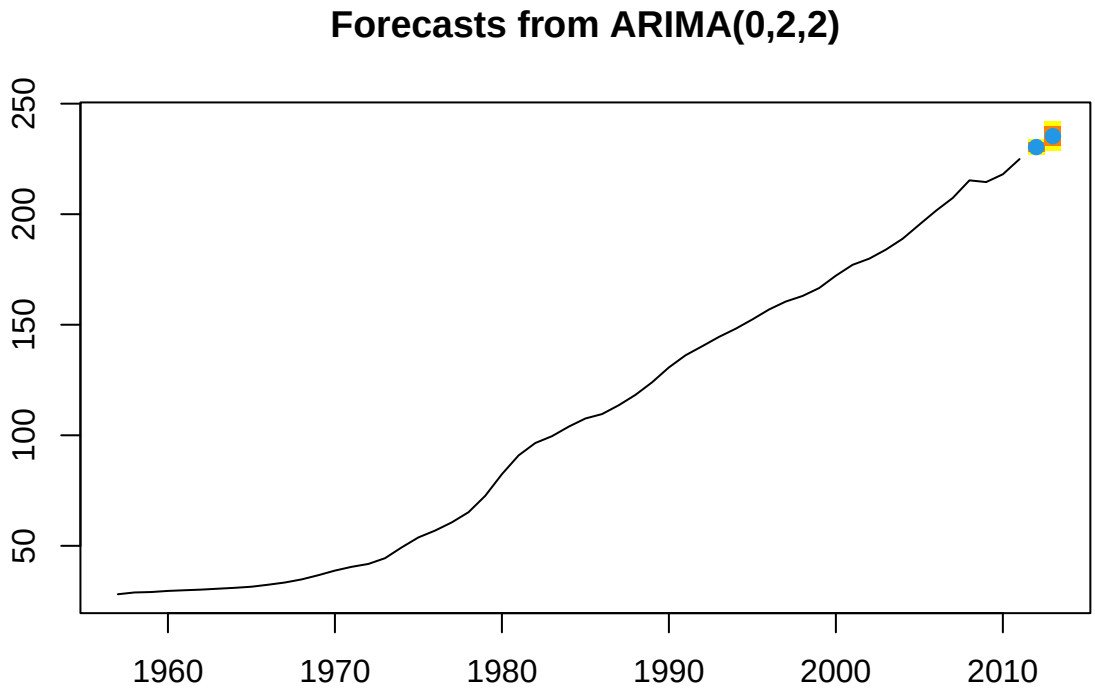
```
plot(forecast(cpi_model,h=1),shadecols="oldstyle")
```



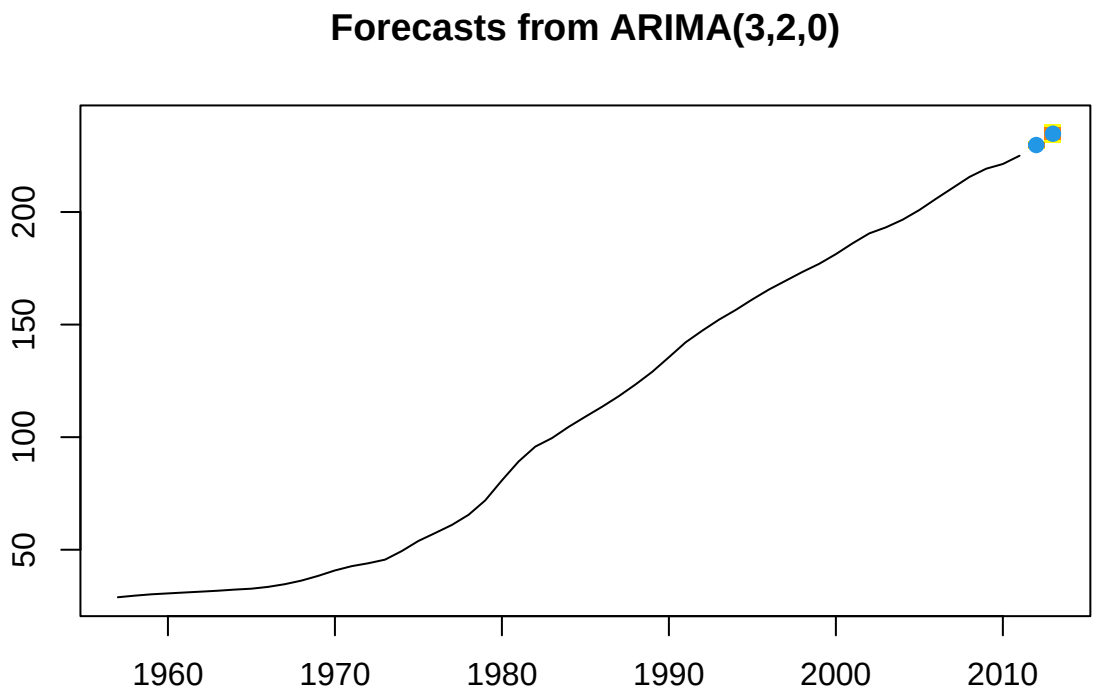

```
plot(forecast(cpi_nofoodenergy_model,h=1),shadecols="oldstyle")
```




```
plot(forecast(cpi_model,h=2),shadecols="oldstyle")
```



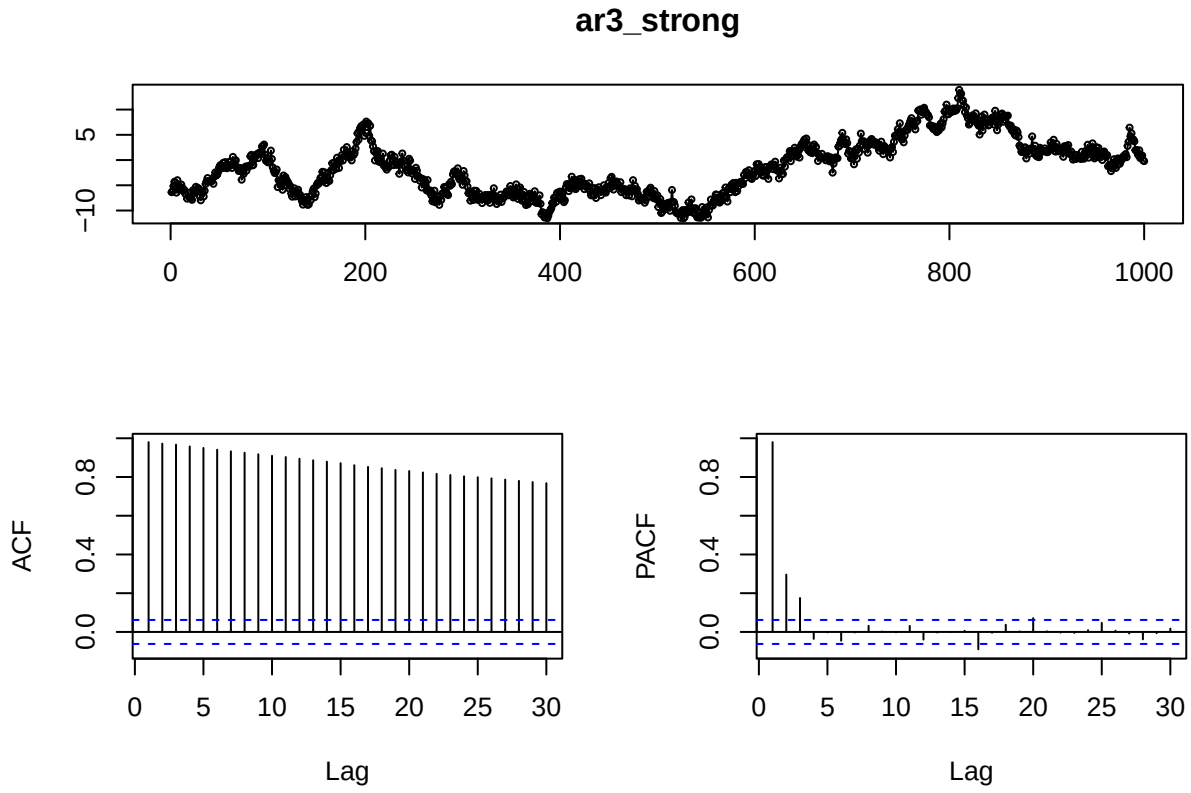

```
plot(forecast(cpi_nofoodenergy_model,h=2),shadecols="oldstyle")
```



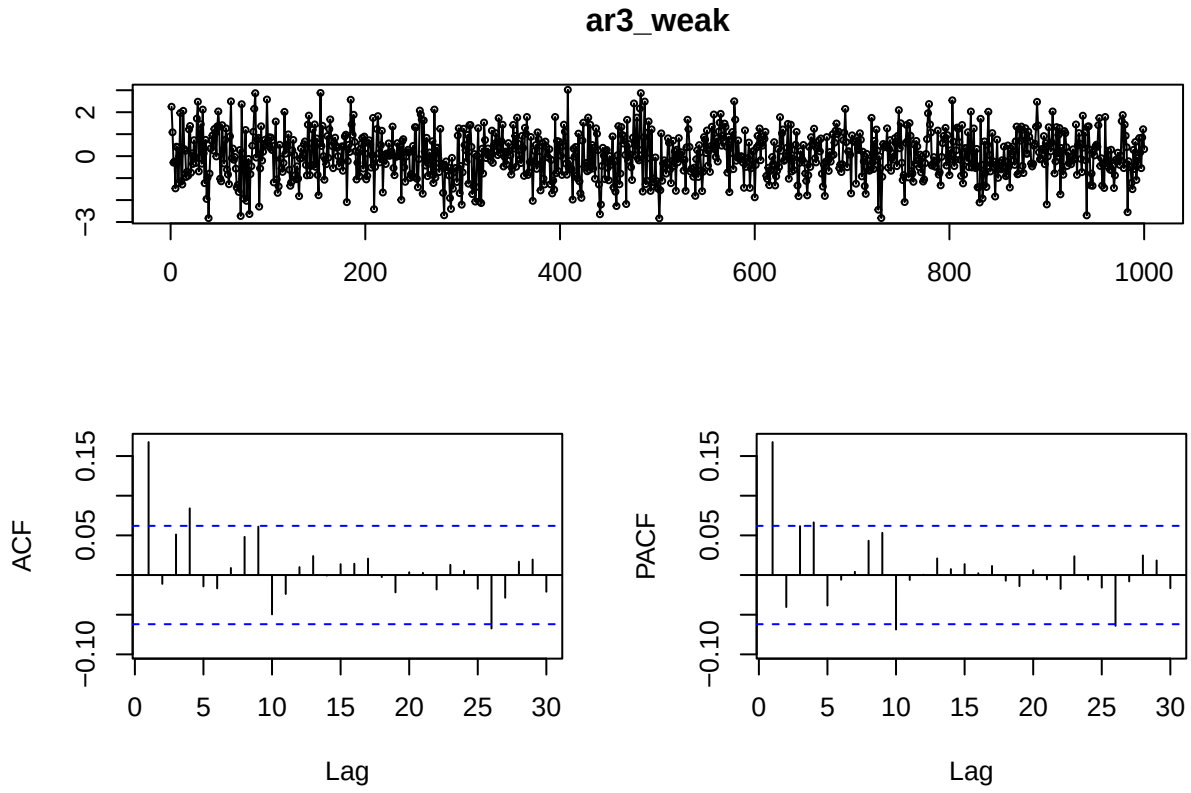
We can see that the variance of the forecasts is lower for the CPI without food and energy. This agrees with the earlier assumption that the CPI without food and energy is easier to predict.

Problem 3

```
set.seed(144)
# Strong persistence AR(3)
ar3_strong <- arima.sim(model = list(ar = c(0.6, 0.25, 0.13)), n = 1000)
tsdisplay(ar3_strong)
```




```r
# Weak persistence AR(3)
ar3_weak <- arima.sim(model = list(ar = c(0.2, -0.05, 0.05)), n = 1000)
tsdisplay(ar3_weak)
```



For the high persistence I chose an AR with parameters .6, .25, and .13 which add up to .98. Since this is very close to 1, it has very strong persistence which can be seen by the data taking a while to mean revert. For the weak persistence, the parameters add up to .2 which means that it is much less persistent and the data mean reverts much faster. It also has a PACF and ACF that are far less "clean" compared to the high persistence.

Problem 4

```
# Daily (includes adjusted close)
daily_google <- tq_get("GOOG", from = "2015-01-01", to = "2025-01-01", get = "stock.prices")
head(daily_google)
```

```
## # A tibble: 6 x 8
##   symbol date        open  high   low close   volume adjusted
##   <chr>  <date>     <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>
## 1 GOOG   2015-01-02  26.4  26.5  26.1  26.2 28951268     25.9
## 2 GOOG   2015-01-05  26.1  26.1  25.6  25.6 41196796     25.4
## 3 GOOG   2015-01-06  25.7  25.7  25.0  25.0 57998800     24.8
## 4 GOOG   2015-01-07  25.3  25.3  24.9  25.0 41301082     24.8
## 5 GOOG   2015-01-08  24.8  25.1  24.5  25.1 67071641     24.8
## 6 GOOG   2015-01-09  25.2  25.2  24.7  24.7 41427428     24.5
```

```
# Monthly: get daily then take last observation per month
monthly_google <- daily_google %>%
  dplyr::mutate(month = lubridate::floor_date(date, "month")) %>%
  dplyr::group_by(month) %>%
```

```
  dplyr::slice_tail(n = 1) %>%
  dplyr::ungroup()
head(monthly_google)
```

```
## # A tibble: 6 x 9
##   symbol date        open  high   low close    volume adjusted month
##   <chr>  <date>     <dbl> <dbl> <dbl> <dbl>     <dbl>    <dbl> <date>
## 1 GOOG   2015-01-30  25.7  26.9  25.7  26.7 112127002     26.4 2015-01-01
## 2 GOOG   2015-02-27  27.6  28.2  27.6  27.8  48203982     27.6 2015-02-01
## 3 GOOG   2015-03-31  27.4  27.7  27.3  27.3  31760961     27.1 2015-03-01
## 4 GOOG   2015-04-30  27.4  27.4  26.8  26.9  41644000     26.6 2015-04-01
## 5 GOOG   2015-05-29  26.9  26.9  26.6  26.6  51948000     26.4 2015-05-01
## 6 GOOG   2015-06-30  26.3  26.3  26.0  26.0  44344000     25.8 2015-06-01
```

```
ret_daily=diff(log(daily_google$adjusted))*100
ret_monthly=diff(log(monthly_google$adjusted))*100

fit_daily=auto.arima(ret_daily)
fit_monthly=auto.arima(ret_monthly)
summary(fit_daily)
```

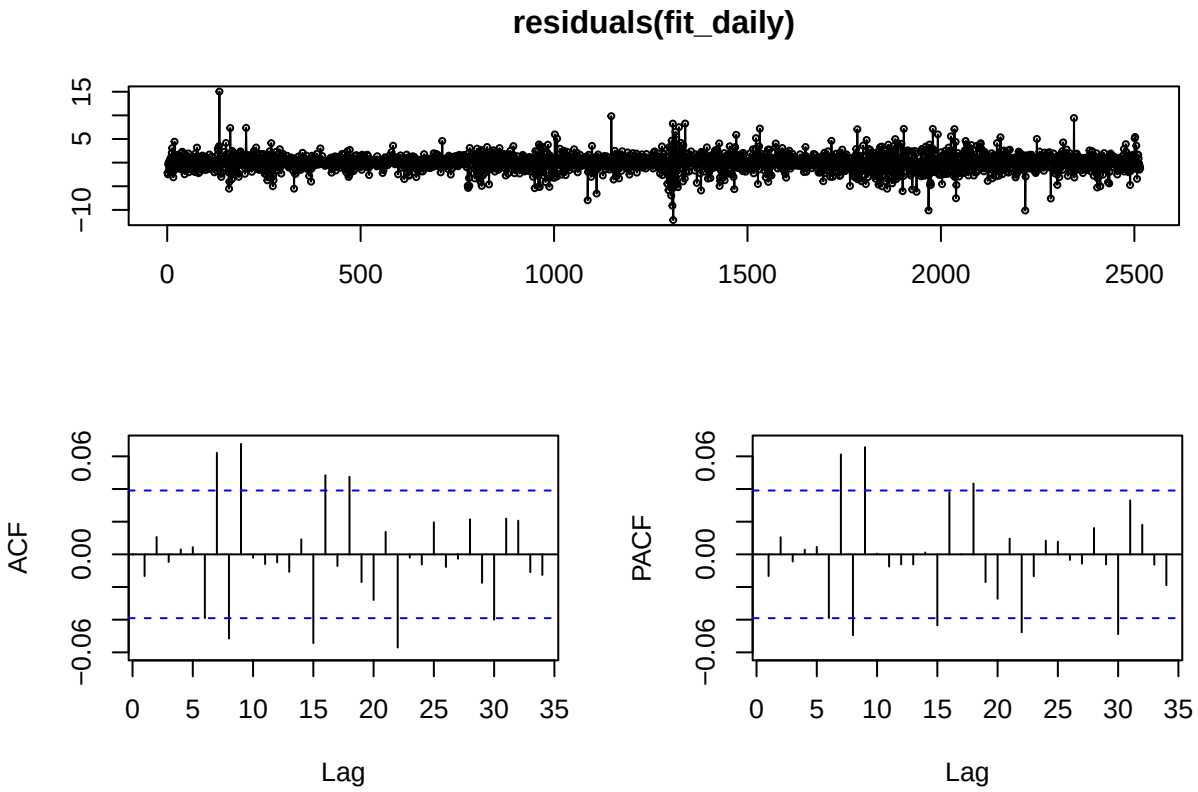
```
## Series: ret_daily
## ARIMA(1,0,1) with non-zero mean
##
## Coefficients:
##          ar1      ma1    mean
##       0.8733  -0.9053  0.0790
## s.e.  0.0918   0.0806  0.0267
##
## sigma^2 = 3.2:  log likelihood = -5029.64
## AIC=10067.28   AICc=10067.29   BIC=10090.6
##
## Training set error measures:
##                        ME    RMSE     MAE      MPE     MAPE      MASE
## Training set 0.0002156881 1.78767 1.22965 100.1078 118.9371 0.6778447
##                     ACF1
## Training set -0.01329957
```

```
summary(fit_monthly)
```

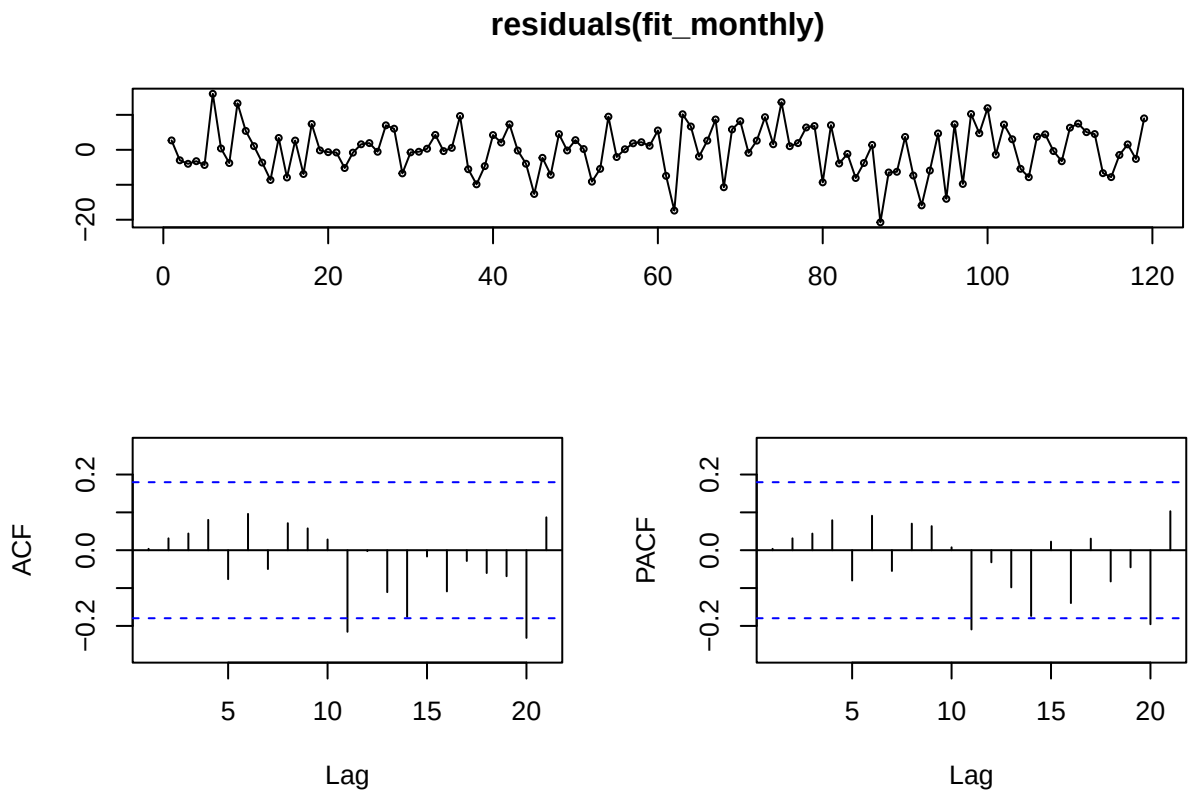
```
## Series: ret_monthly
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##          ar1    mean
##       -0.1913  1.6389
## s.e.   0.0904  0.5134
##
## sigma^2 = 45.15:  log likelihood = -394.56
## AIC=795.12   AICc=795.33   BIC=803.46
##
```

```
## Training set error measures:
##                       ME     RMSE      MAE      MPE     MAPE      MASE
## Training set 0.003967166 6.662721 5.26966 107.8241 127.4841 0.6401926
##                     ACF1
## Training set 0.003712685
```

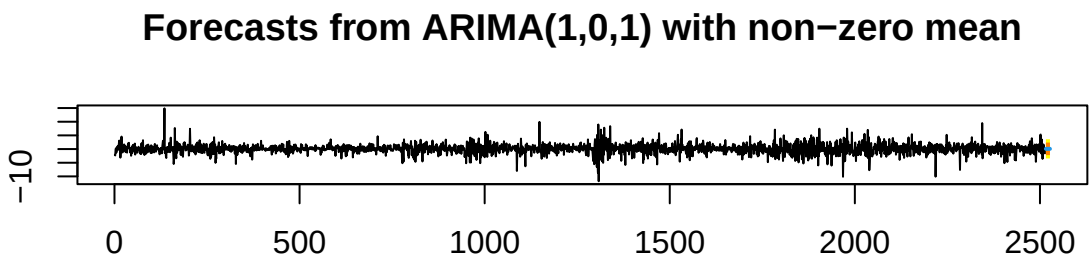
```
tsdisplay(residuals(fit_daily))
```

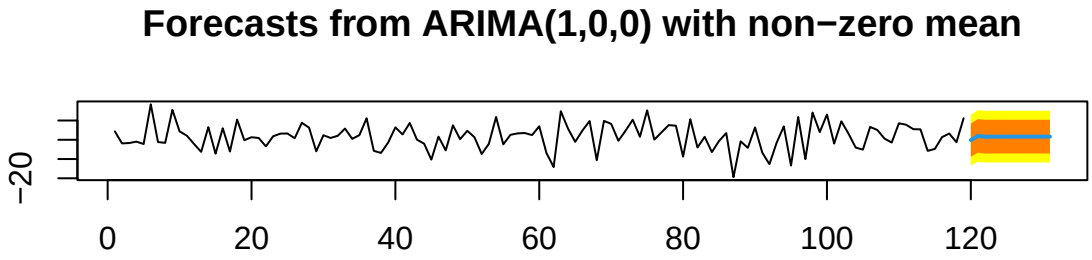



```
tsdisplay(residuals(fit_monthly))
```




```
par(mfrow=c(2,1))
plot(forecast(fit_daily,h=12),shadecols="oldstyle")
plot(forecast(fit_monthly,h=12),shadecols="oldstyle")
```





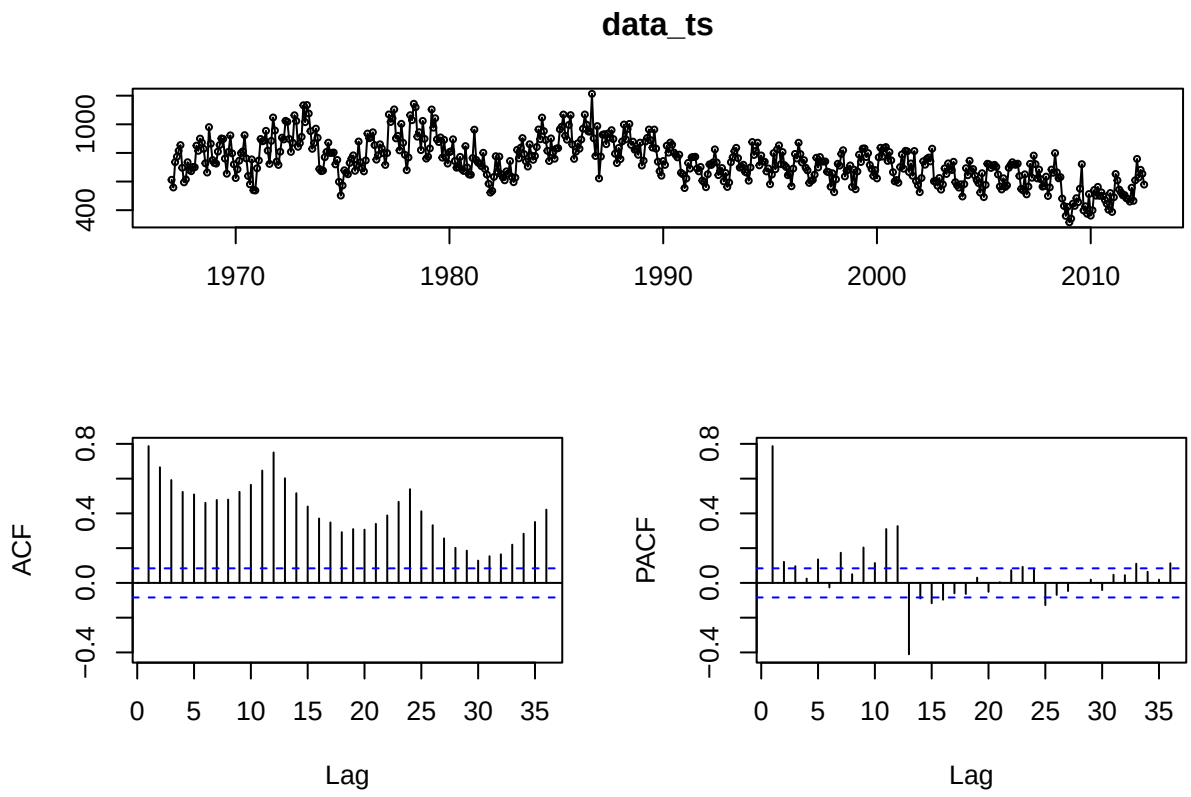
The forecasts stay close to the mean, however, the density of the forecasts shows that they are extremely volatile.

Problem 5

```
data=read_excel("../../data/Chapter8_Exercises_Data.xlsx", sheet="Exercise 10")
data_ts=ts(data[4], start = c((1967), 1), frequency = 12)
head(data_ts)
```

```
##        Jan   Feb   Mar   Apr   May   Jun
## 1967 612.2 558.5 735.8 773.6 814.9 854.6
```

```
tsdisplay(data_ts)
```




```
fit_data=auto.arima(data_ts)
summary(fit_data)
```

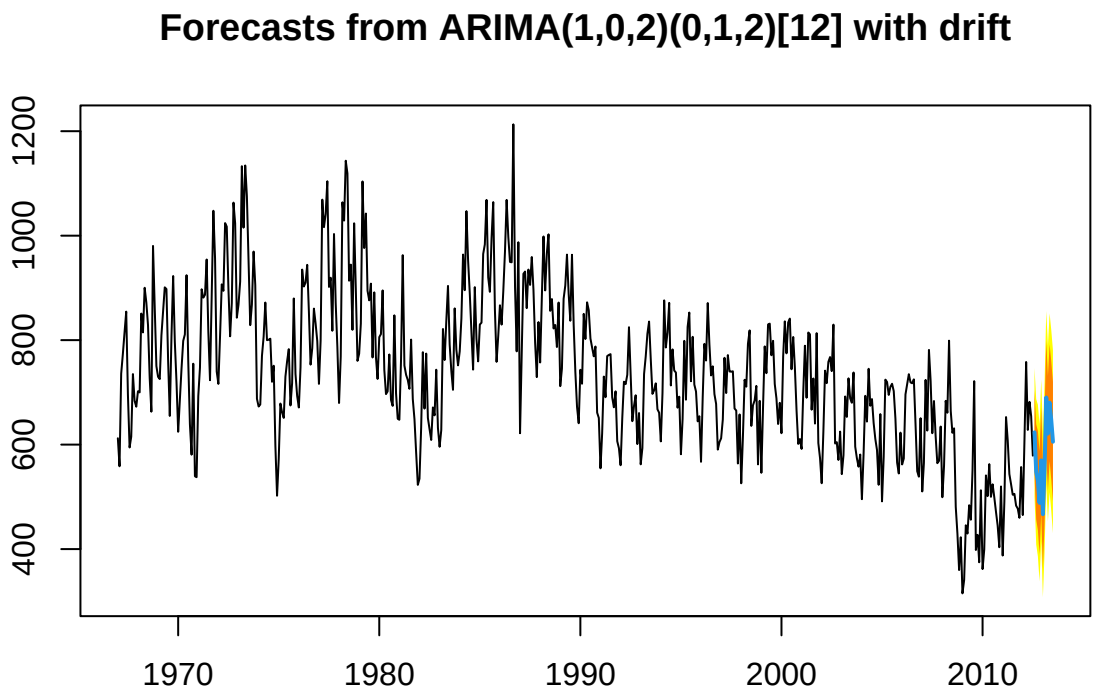
```
## Series: data_ts
## ARIMA(1,0,2)(0,1,2)[12] with drift
##
## Coefficients:
##          ar1      ma1      ma2     sma1     sma2    drift
##       0.9506  -0.4267  -0.1622  -0.6676  -0.1655  -0.3915
## s.e.  0.0183   0.0471   0.0444   0.0431   0.0415   0.3530
##
## sigma^2 = 3957:  log likelihood = -2978.38
## AIC=5970.76   AICc=5970.97   BIC=6000.73
##
## Training set error measures:
##                      ME     RMSE      MAE        MPE     MAPE      MASE
## Training set 0.9180193 61.86296 45.61878 -0.4773104 6.300844 0.6159717
##                      ACF1
## Training set -0.004087038
```

```
summary(forecast(fit_data,h=12))
```

```
##
## Forecast method: ARIMA(1,0,2)(0,1,2)[12] with drift
##
```

```
## Model Information:
## Series: data_ts
## ARIMA(1,0,2)(0,1,2)[12] with drift
##
## Coefficients:
##          ar1      ma1      ma2     sma1     sma2    drift
##       0.9506  -0.4267  -0.1622  -0.6676  -0.1655  -0.3915
## s.e.  0.0183   0.0471   0.0444   0.0431   0.0415   0.3530
##
## sigma^2 = 3957:  log likelihood = -2978.38
## AIC=5970.76   AICc=5970.97   BIC=6000.73
##
## Error measures:
##                       ME     RMSE      MAE        MPE     MAPE      MASE
## Training set 0.9180193 61.86296 45.61878 -0.4773104 6.300844 0.6159717
##                       ACF1
## Training set -0.004087038
##
## Forecasts:
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Aug 2012       623.5021 542.8840 704.1202 500.2073 746.7969
## Sep 2012       547.9390 456.9254 638.9527 408.7457 687.1324
## Oct 2012       531.2318 436.2757 626.1878 386.0090 676.4545
## Nov 2012       488.5639 390.1810 586.9468 338.1003 639.0276
## Dec 2012       569.6462 468.2661 671.0262 414.5988 724.6936
## Jan 2013       466.8699 362.8557 570.8841 307.7939 625.9459
## Feb 2013       555.8701 449.5316 662.2087 393.2394 718.5009
## Mar 2013       690.1862 581.7901 798.5822 524.4087 855.9636
## Apr 2013       621.1895 510.9672 731.4118 452.6190 789.7600
## May 2013       680.1659 568.3189 792.0129 509.1106 851.2212
## Jun 2013       646.2146 532.9195 759.5098 472.9446 819.4846
## Jul 2013       605.4981 490.9100 720.0861 430.2508 780.7454
```

```
plot(forecast(fit_data,h=12),shadecols="oldstyle")
```

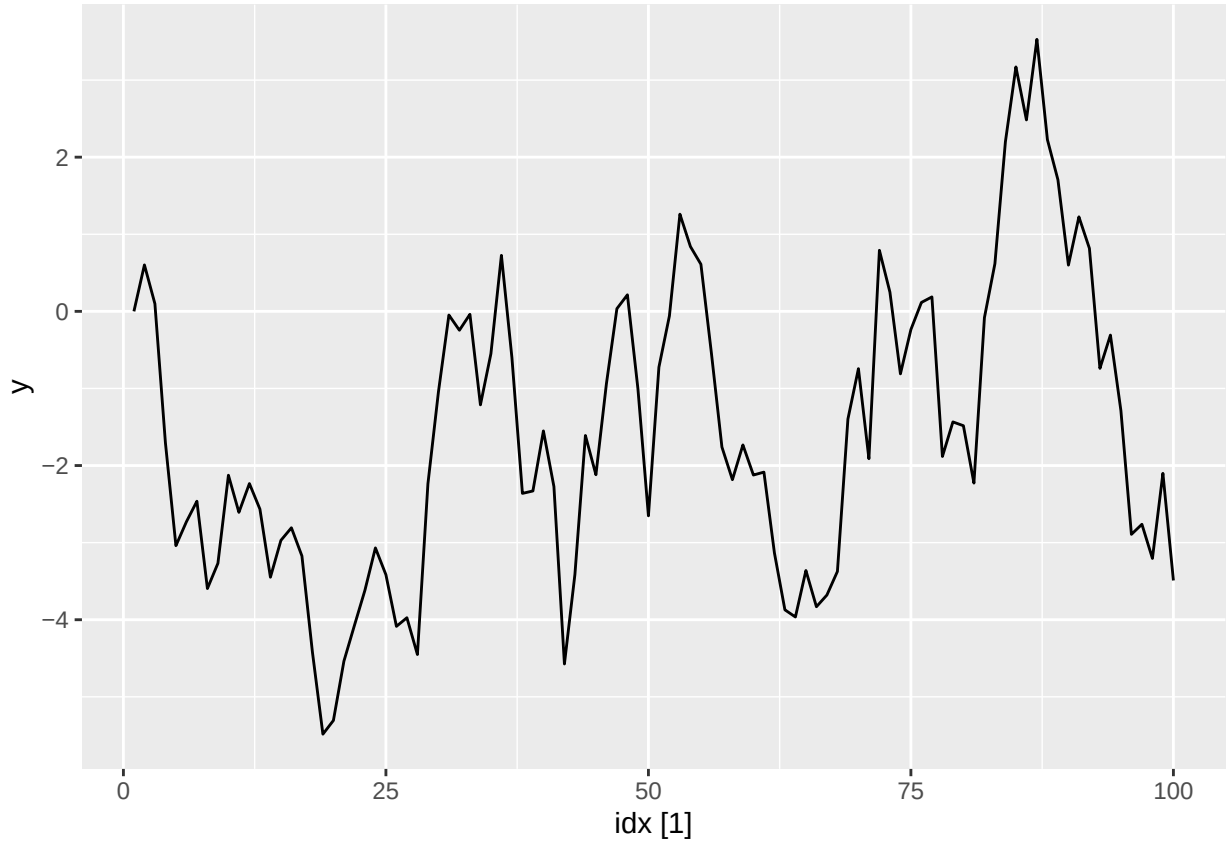


From auto.arima, the best model has an ARMA(1,2) with a seasonal MA(2) and one seasonal difference, with a frequency of 12. This indicates the presence of seasonal dynamics in the data. The forecast shows this seasonal component.

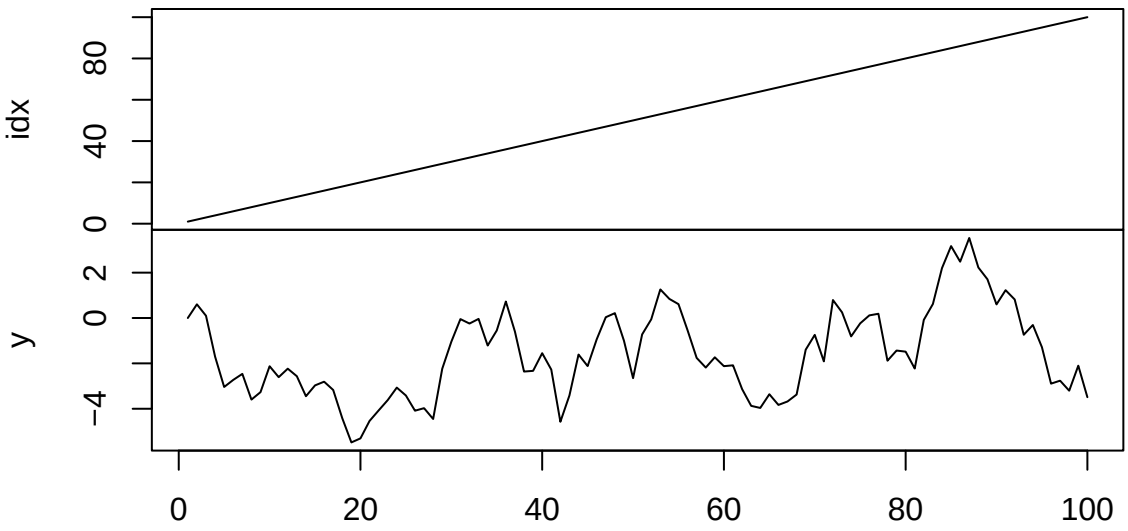
Problem 6

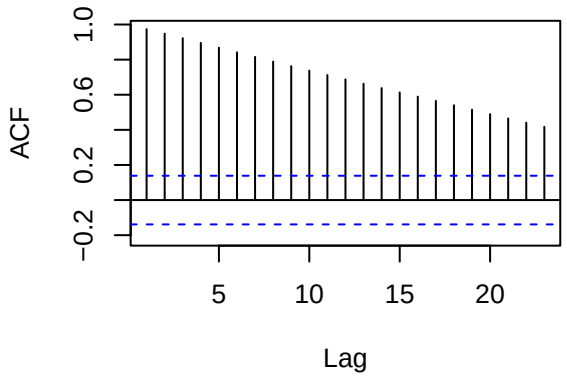
```
set.seed(144)
y <- numeric(100)
e <- rnorm(100)
for(i in 2:100)
  y[i] <- 0.95*y[i-1] + e[i]
sim <- tsibble(idx = seq_len(100), y = y, index = idx)

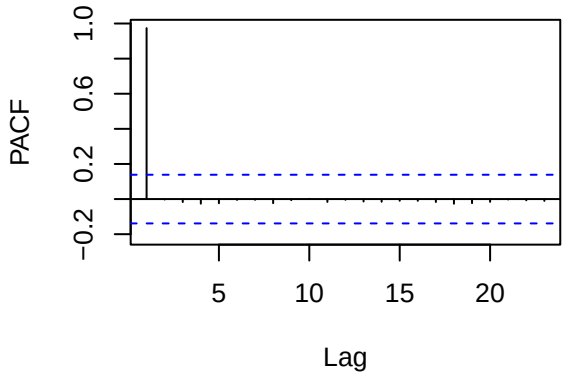
autoplot(sim)
```




```
tsdisplay(sim)
```


sim
idx
y
0
40
80
−4
0
2
0
20
40
60
80
100





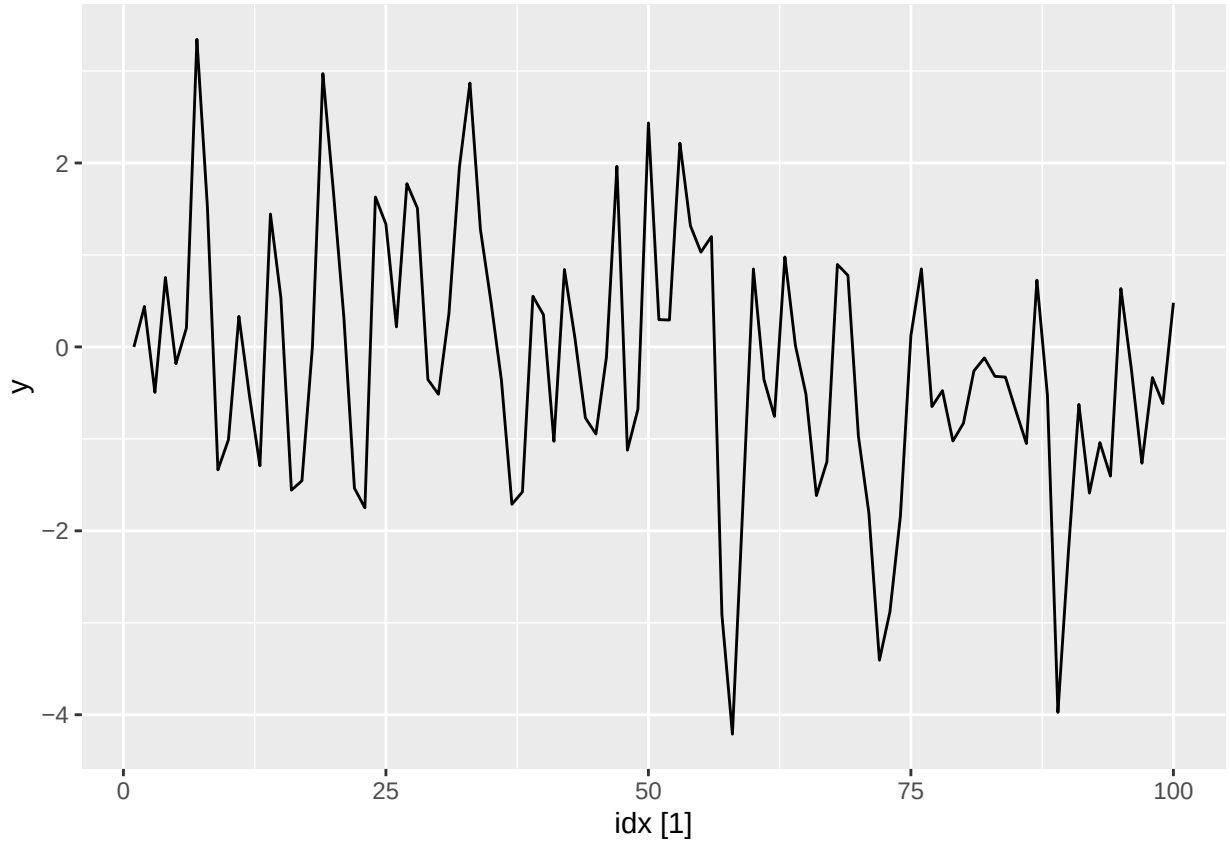
b) As you increase phi, the data becomes more persistent.

```
e <- rnorm(100) # White noise shocks
y_ma <- numeric(100)

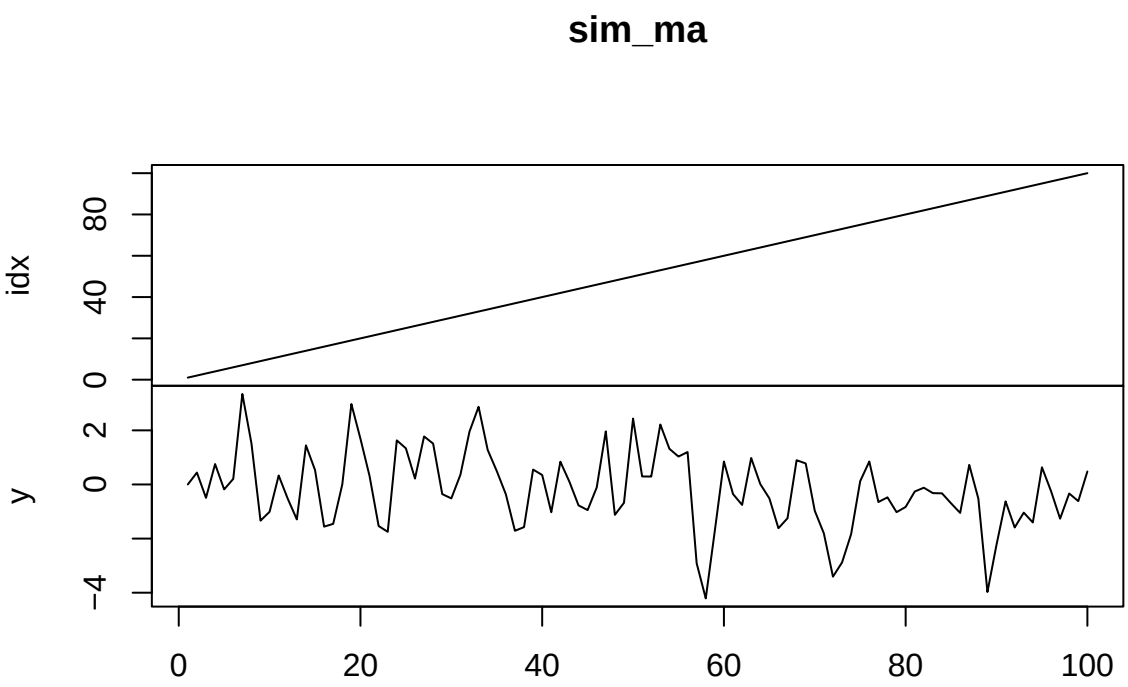
for(i in 2:100) {
  y_ma[i] <- e[i] + 0.999 * e[i-1] # Note: current error + weighted past error
}

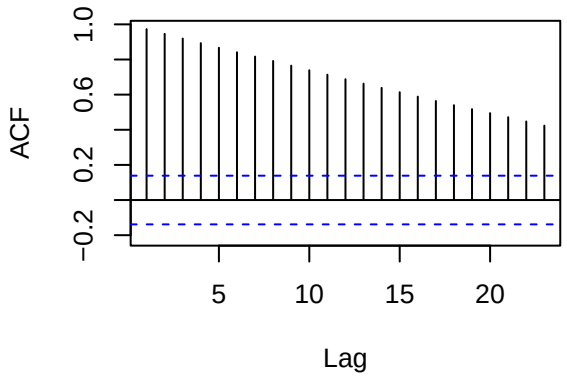
sim_ma <- tsibble(idx = seq_len(100), y = y_ma, index = idx)

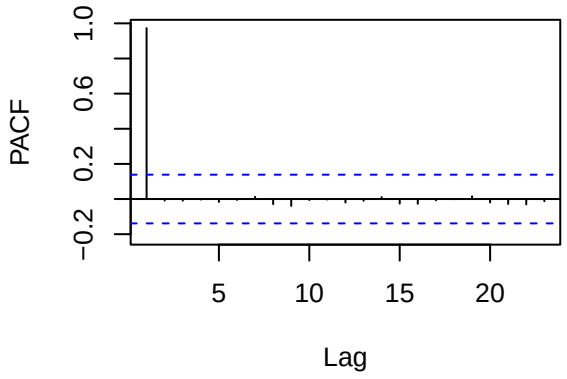
# Part d: Plot
autoplot(sim_ma)
```




```
tsdisplay(sim_ma)
```


sim_ma
idx
y
0
40
80
0
2
−4
0
20
40
60
80
100





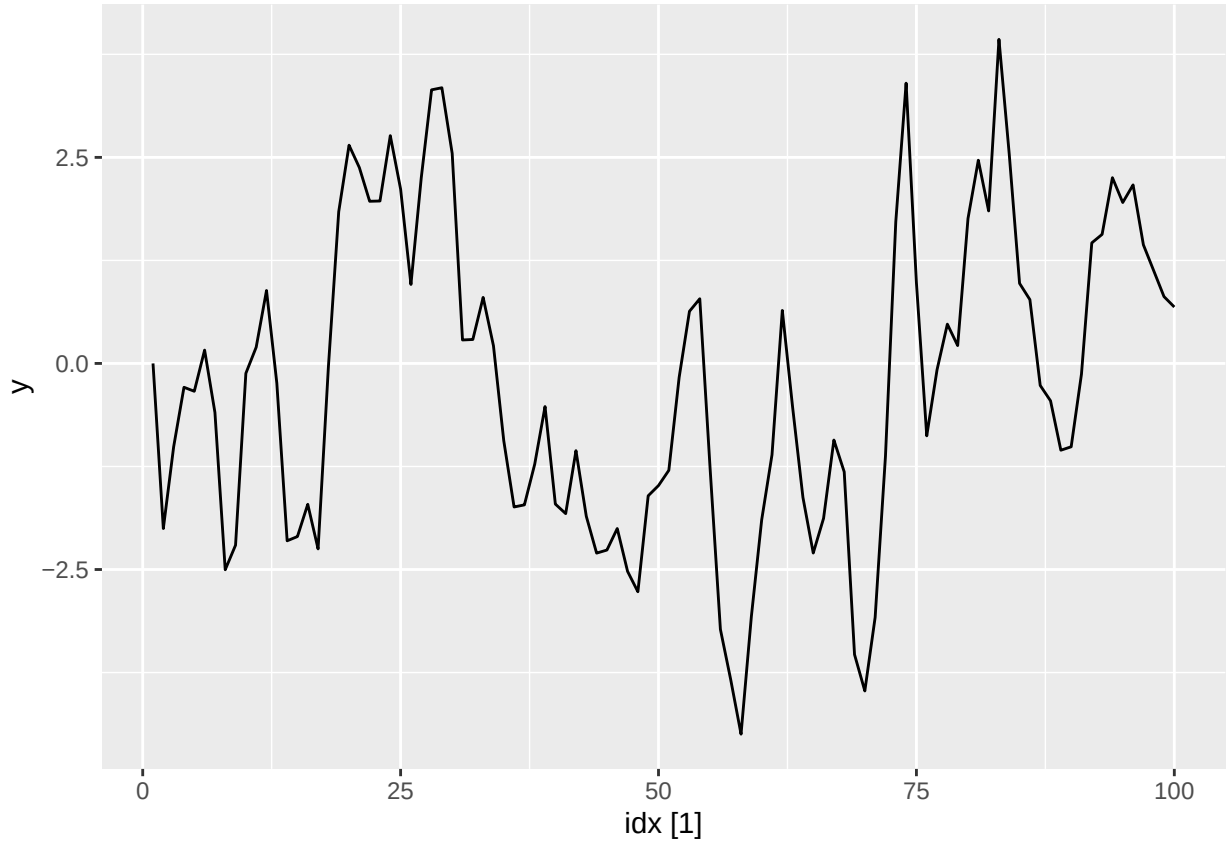
As you increase the value of theta, the data becomes slightly smoother and more persistent.

```
y_arma <- numeric(100)
e <- rnorm(100)

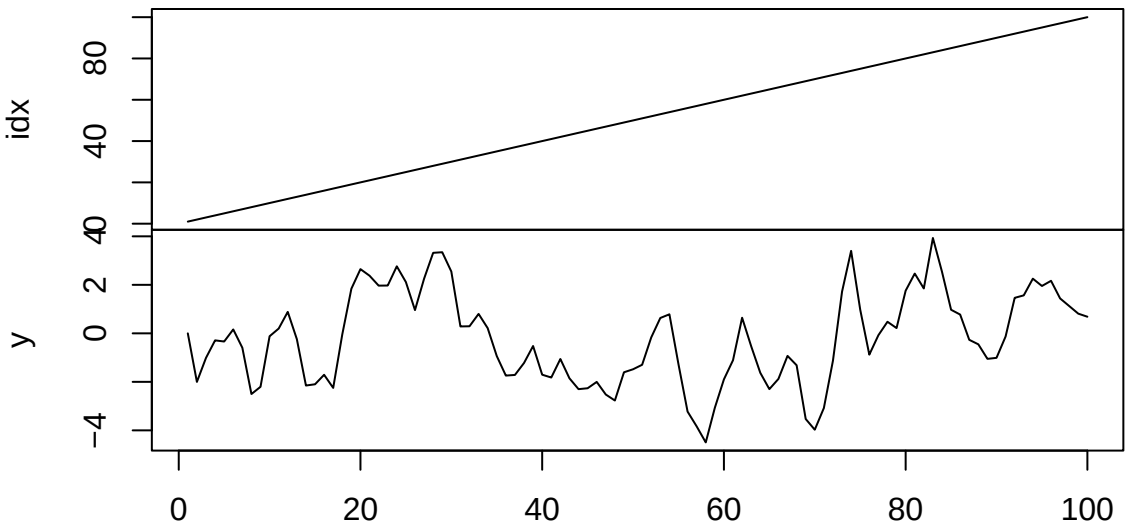
for(i in 2:100) {
  y_arma[i] <- 0.6 * y_arma[i-1] + e[i] + 0.6 * e[i-1]
}

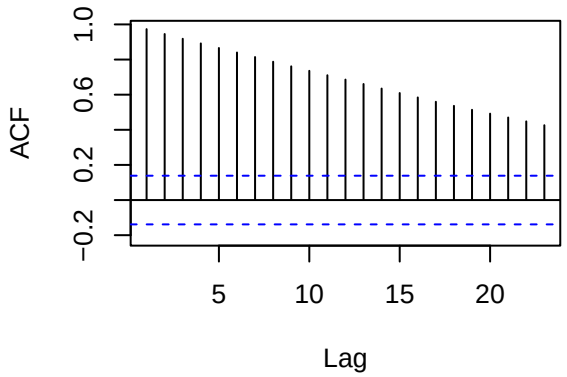
sim_arma <- tsibble(idx = seq_len(100), y = y_arma, index = idx)

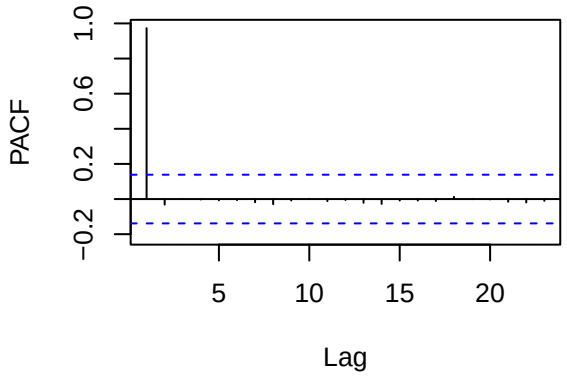
autoplot(sim_arma)
```




```
tsdisplay(sim_arma)
```


sim_arma
idx
y
80
40
40
2
0
−4
0
20
40
60
80
100



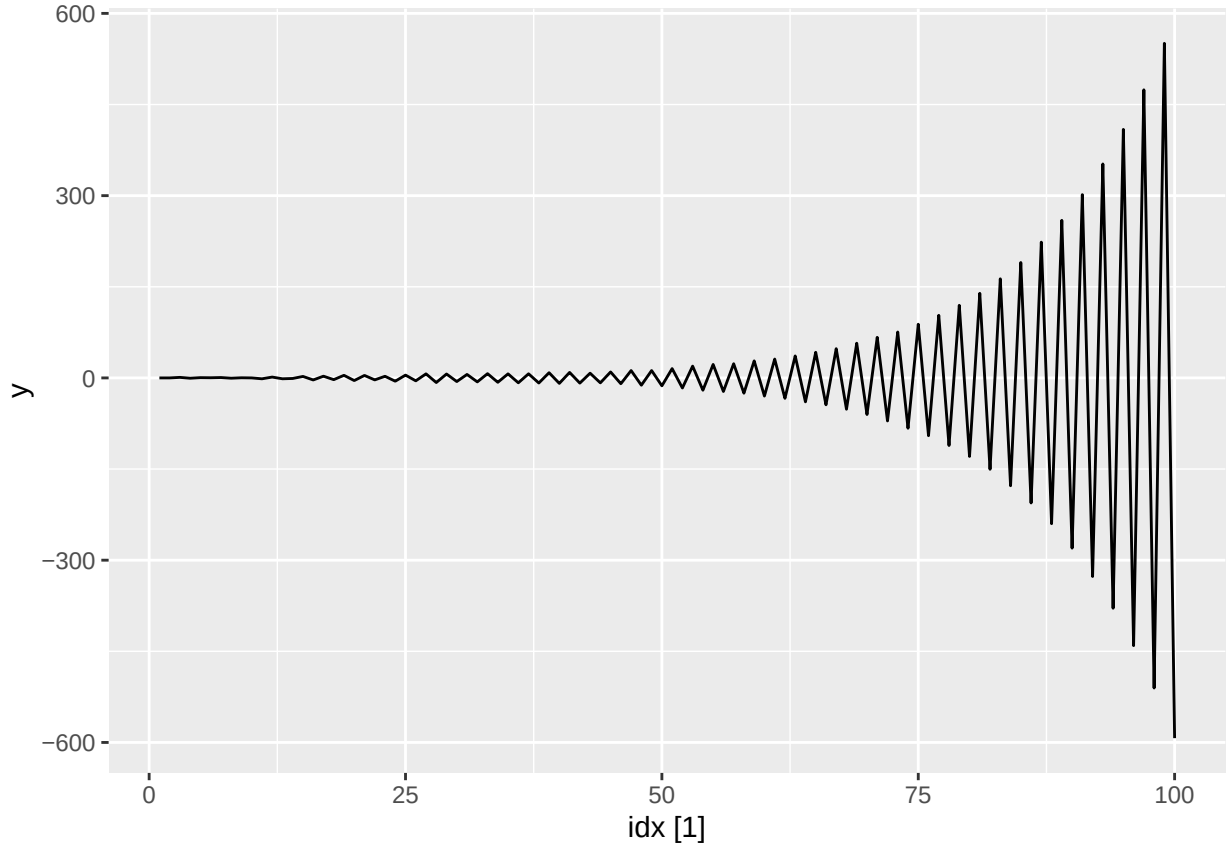



```
y_ar2 <- numeric(100)
e <- rnorm(100)

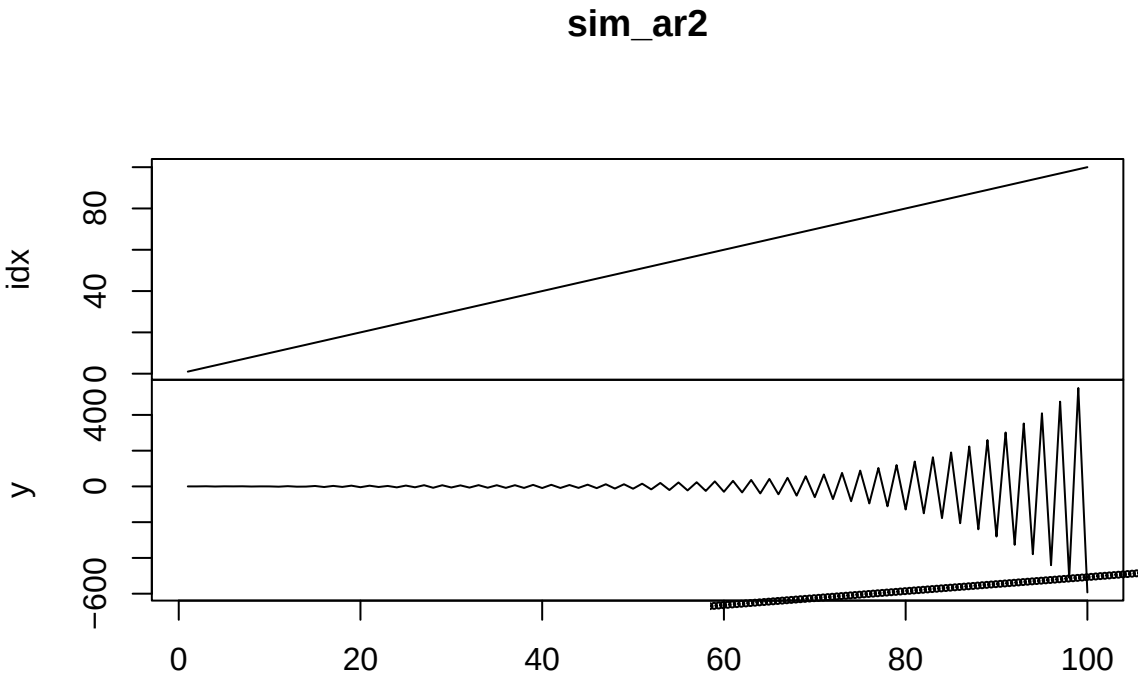
for(i in 3:100) {
  y_ar2[i] <- (-0.8 * y_ar2[i-1]) + (0.3 * y_ar2[i-2]) + e[i]
}

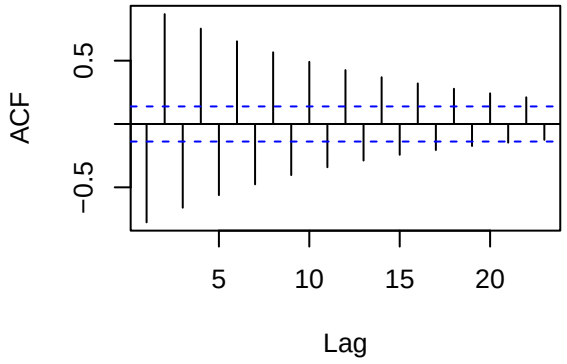
sim_ar2 <- tsibble(idx = seq_len(100), y = y_ar2, index = idx)

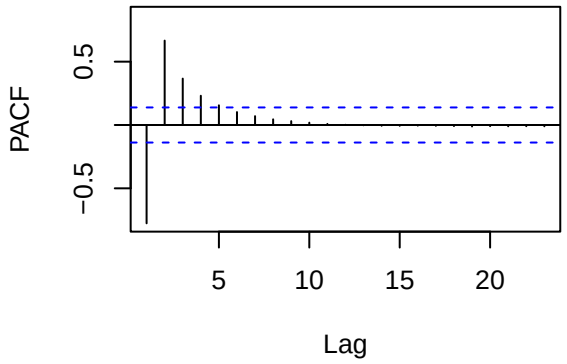
autoplot(sim_ar2)
```




```
tsdisplay(sim_ar2)
```


sim_ar2
idx
y
80
40
0
4000
0
−600
0
20
40
60
80
100





The ARMA(1,1) does not look to have clear pattern just from looking at it and has confusing mix of persistence and mean reversion. The AR(2) is divergent because one root of its characteristic polynomial lies inside the unit circle (phi2 - phi1 = 1.1 > 1), so the process is not stationary.

Problem 7

```
fit=auto.arima(aus_airpassengers)
head(aus_airpassengers)
```

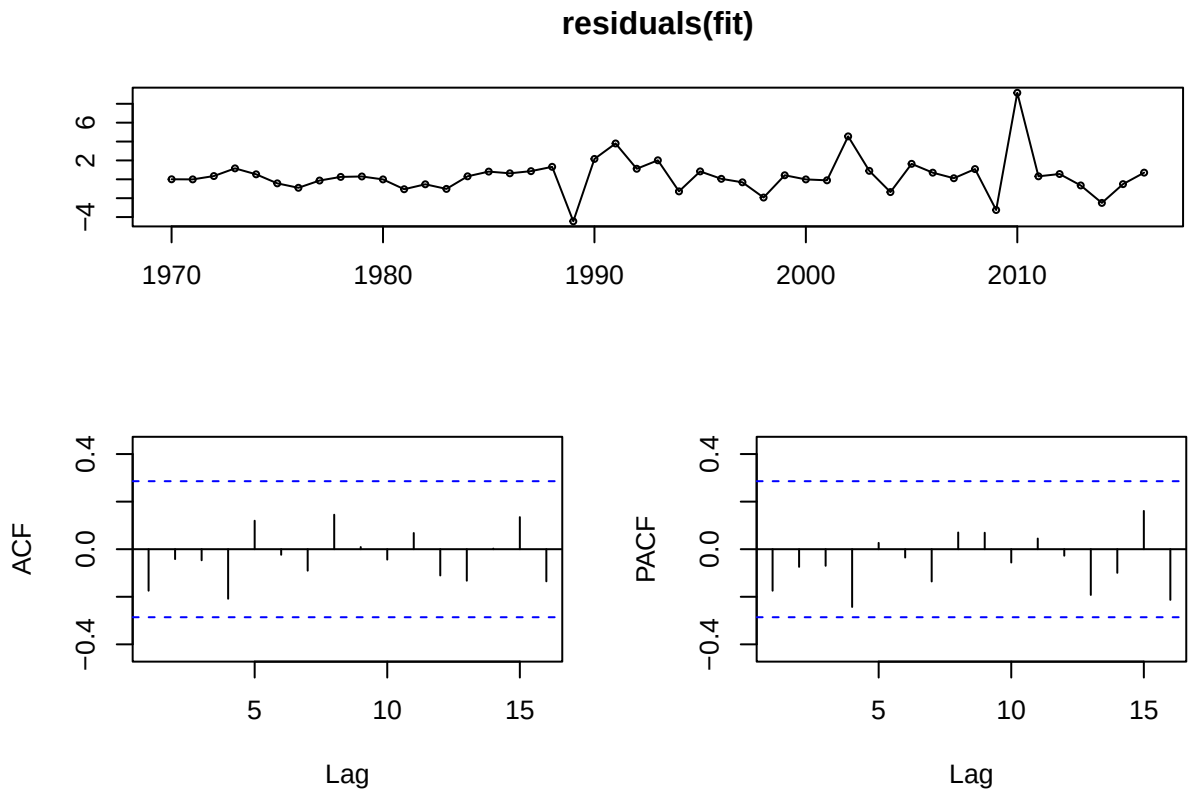
```
## # A tsibble: 6 x 2 [1Y]
##    Year Passengers
##   <dbl>      <dbl>
## 1  1970       7.32
## 2  1971       7.33
## 3  1972       7.80
## 4  1973       9.38
## 5  1974      10.7
## 6  1975      11.1
```

```
summary(fit)
```

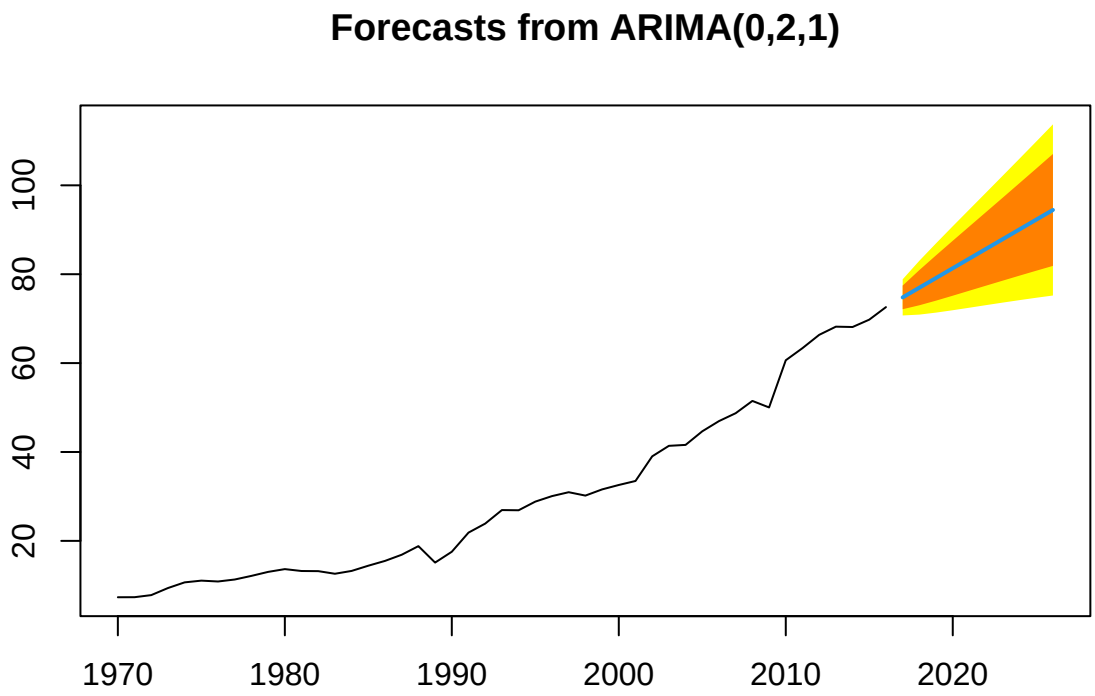
```
## Series: aus_airpassengers
## ARIMA(0,2,1)
##
## Coefficients:
```

```
##           ma1
##       -0.8963
## s.e.   0.0594
##
## sigma^2 = 4.308:  log likelihood = -97.02
## AIC=198.04   AICc=198.32   BIC=201.65
##
## Training set error measures:
##                      ME     RMSE      MAE       MPE     MAPE     MASE       ACF1
## Training set 0.3453292 2.008183 1.215008 0.9528709 4.809966 0.699953 -0.1743841
```

```
tsdisplay(residuals(fit))
```




```
plot(forecast(fit,h=10),shadecols="oldstyle")
```



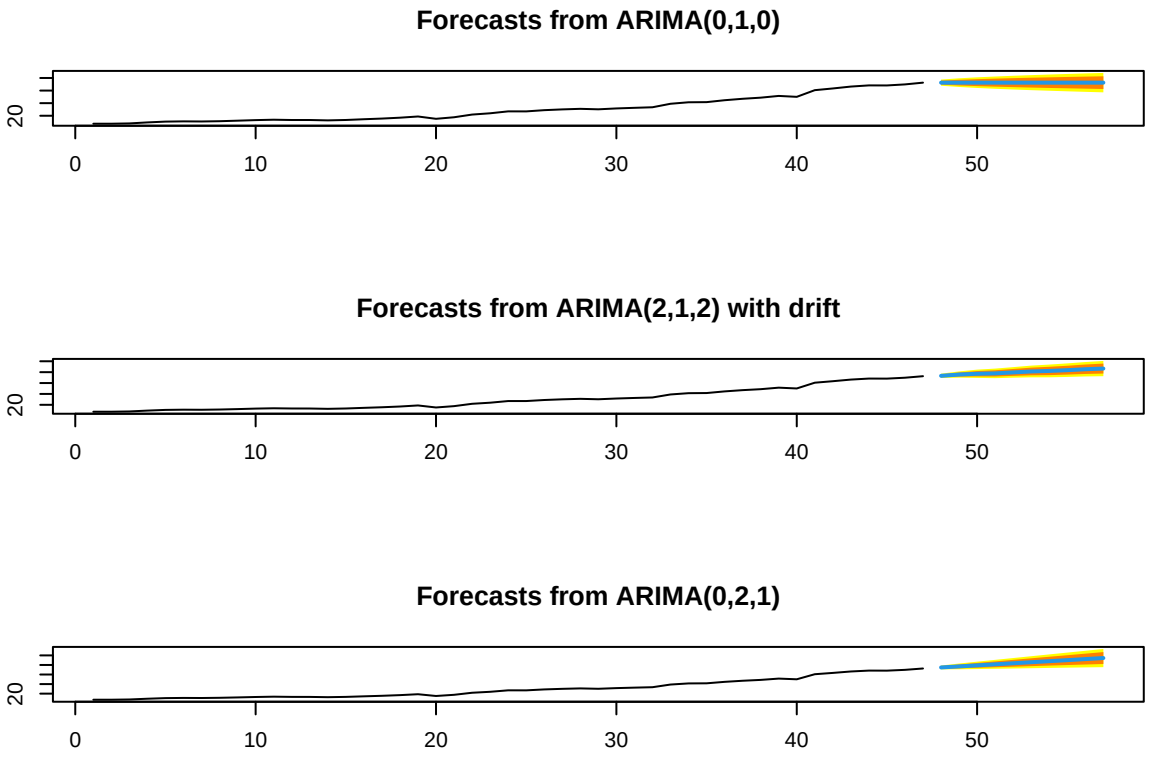
Auto Arima chose an ARIMA(0,2,1) which is an MA1 that has been differenced twice.

b)

(1-B)^2 y_t = (1-.8963B)e_t (there is no constant because the series is differenced twice).

c,d,e)

```
arma010<-arima(aus_airpassengers$Passengers, order = c(0, 1, 0))
arma212 <- Arima(aus_airpassengers$Passengers, order = c(2, 1, 2), include.drift = TRUE)
arma021<-arima(aus_airpassengers$Passengers, order = c(0, 2, 1))
par(mfrow=c(3,1))
plot(forecast(arma010,h=10),shadecols="oldstyle")
plot(forecast(arma212,h=10),shadecols="oldstyle")
plot(forecast(arma021,h=10),shadecols="oldstyle")
```

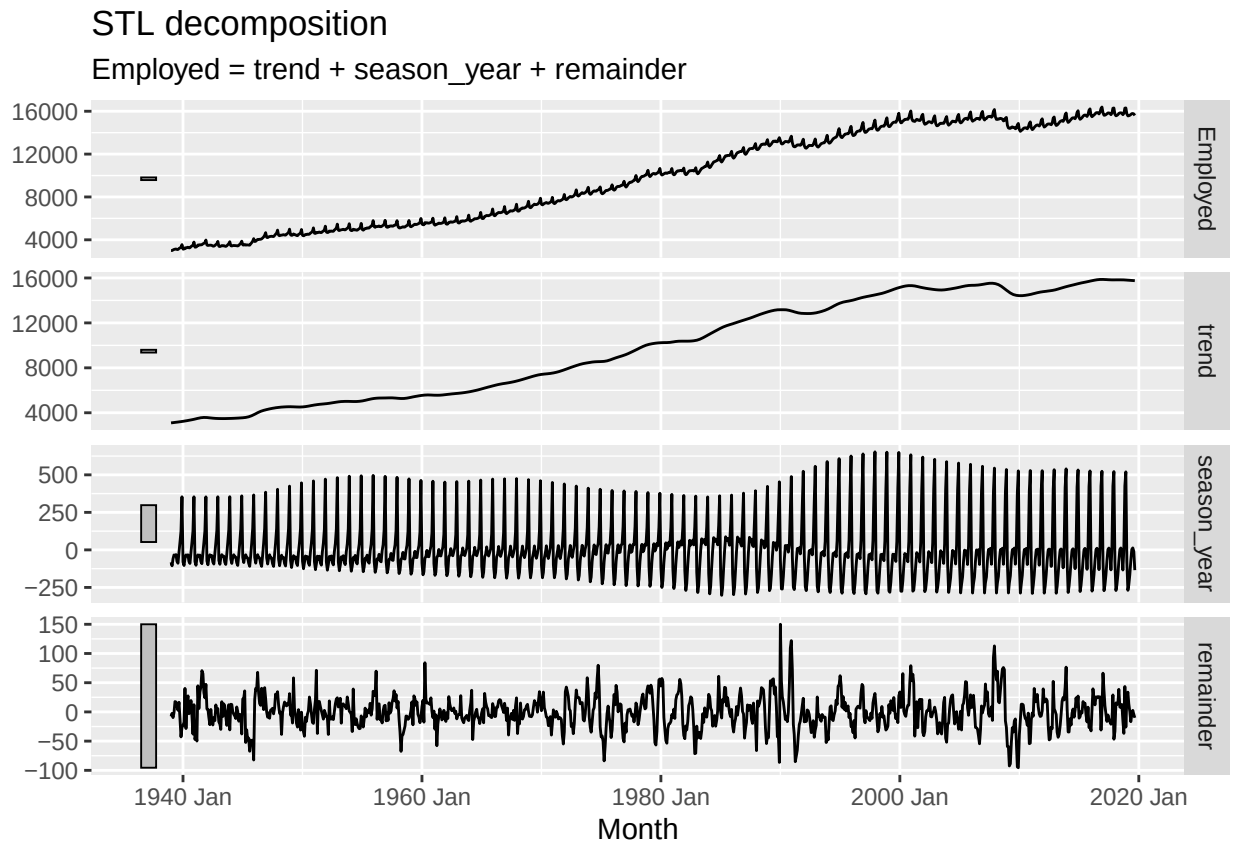


The arima(0,1,0) has no drift, so its forecast is flat compared to the model in part a. If you try to include a constant in the arma021, there is an error because the model is not stationary. In part d, if you remove the constant there is an error and the model is not able to compute.

Problem 8

```
employment_subset <- us_employment %>%
  filter(Title == "Retail Trade")

employment_subset %>%
  model(STL(Employed)) %>%
  components() %>%
  autoplot()
```

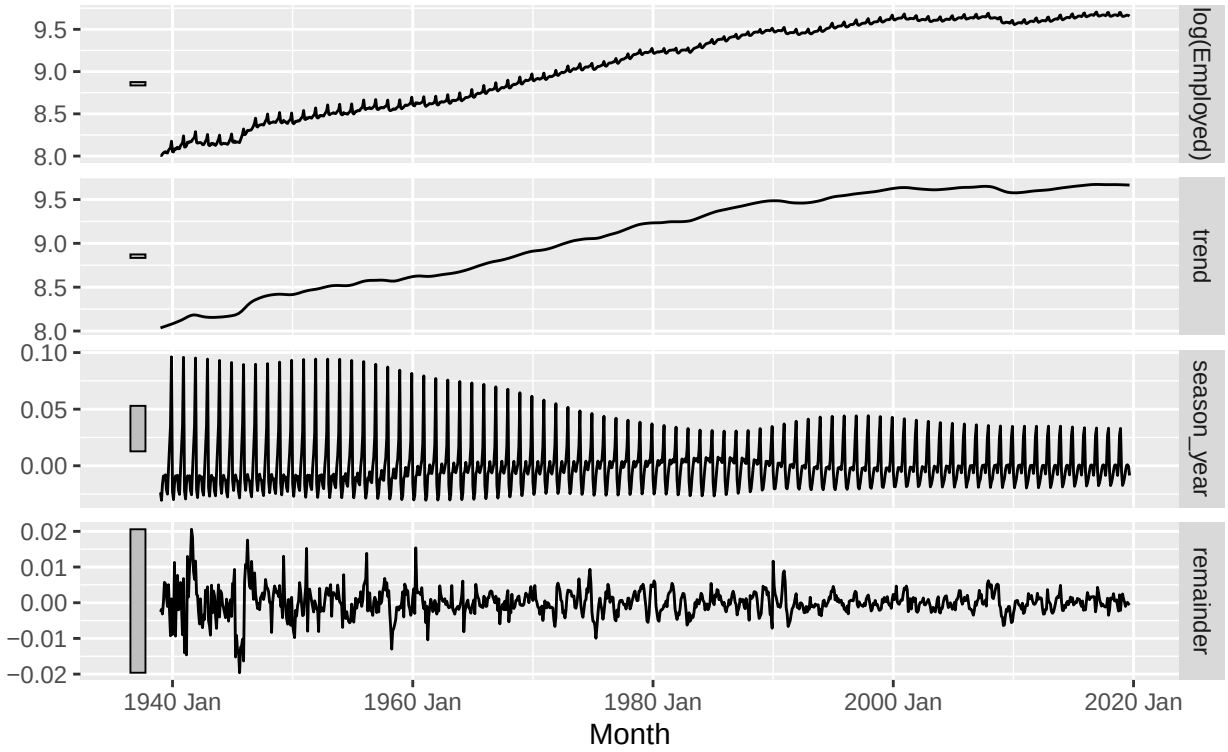


The trend is upward and positive and the seasonal component is very strong but highly volatile.

b)

```
retail_log <- us_employment %>%
  filter(Title == "Retail Trade")

retail_log %>%
  model(STL(log(Employed))) %>%
  components() %>%
  autoplot()
```



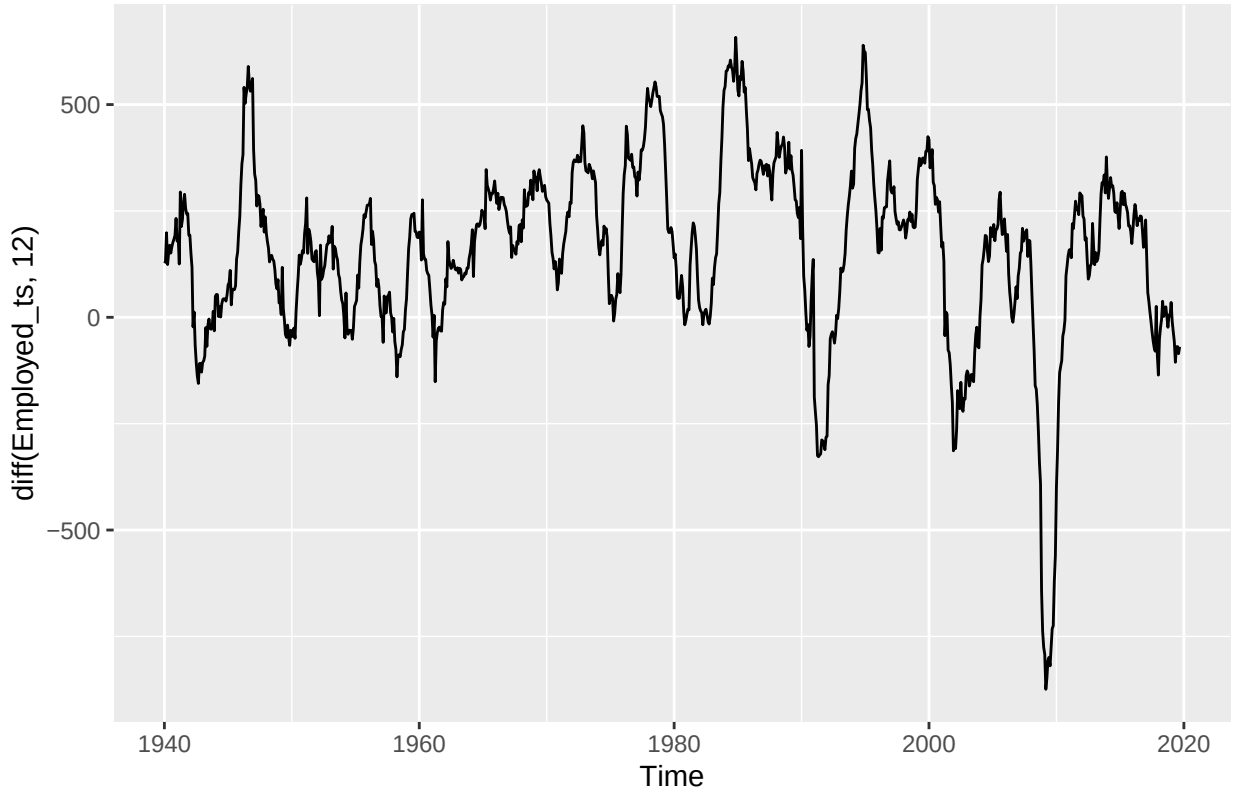
I applied a log transformation to the data.

c)

No, the data is not stationary.

```
Employed <- employment_subset$Employed
Employed_ts <- ts(Employed, start = c(1939, 1), frequency = 12)

autoplot(diff(Employed_ts, 12))
```

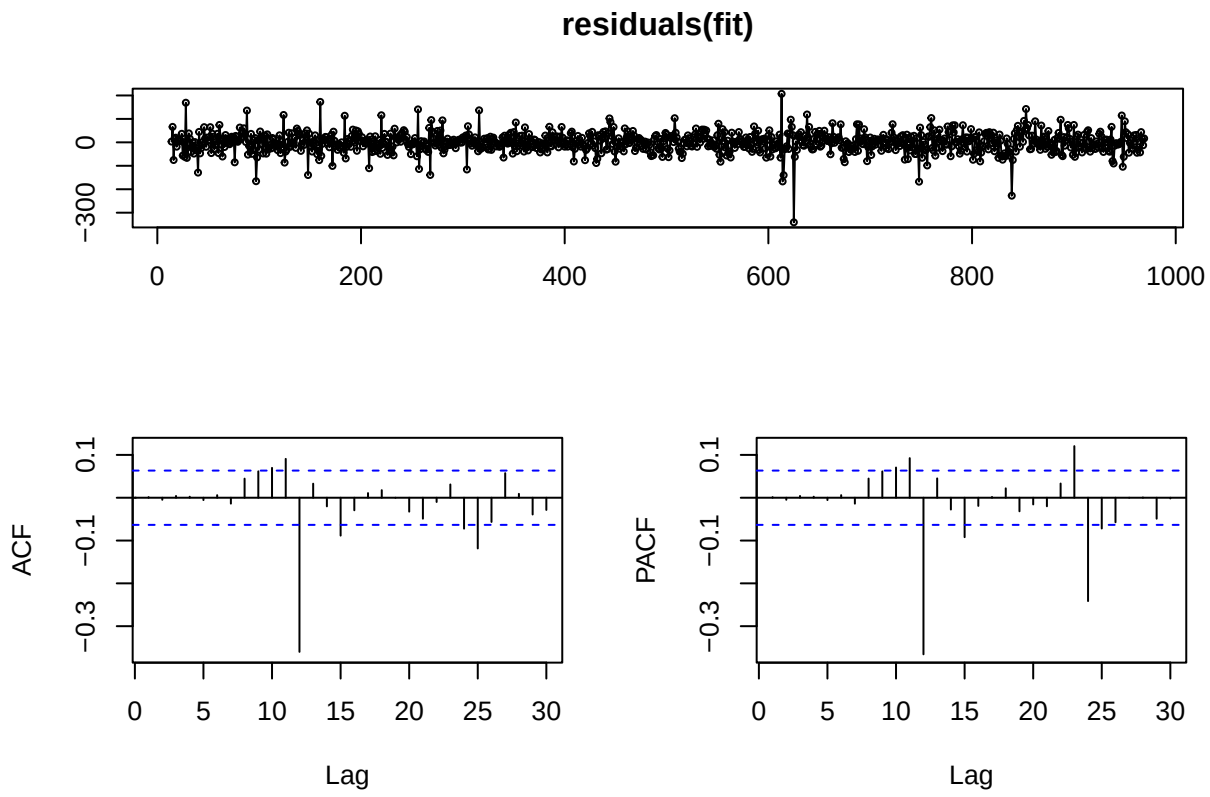


The data is now stationary.

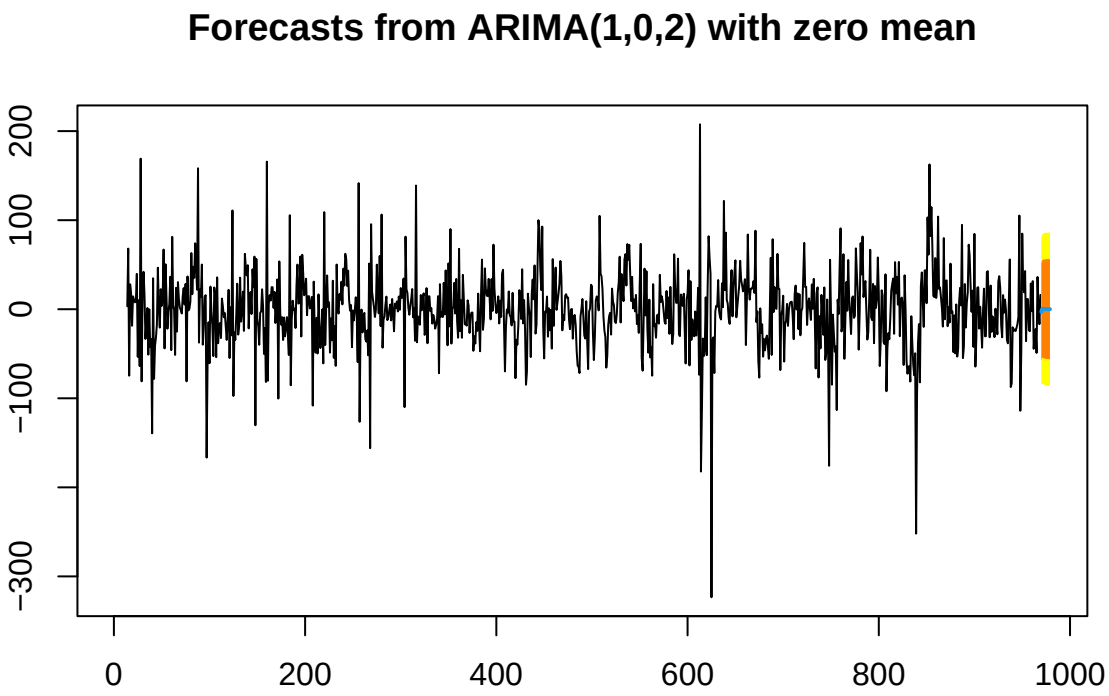
```
fit=auto.arima(difference(Employed, 12) %>% difference())
summary(fit)
```

```
## Series: difference(Employed, 12) %>% difference()
## ARIMA(1,0,2) with zero mean
##
## Coefficients:
##          ar1      ma1     ma2
##       0.7984  -0.8217  0.1720
## s.e.  0.0509   0.0582  0.0322
##
## sigma^2 = 1812:  log likelihood = -4941.22
## AIC=9890.43   AICc=9890.47   BIC=9909.88
##
## Training set error measures:
##                       ME     RMSE      MAE       MPE     MAPE      MASE
## Training set -0.1242186 42.50446 30.08808 -69.38171 388.7011 0.7208356
##                     ACF1
## Training set 0.00139614
```

```
tsdisplay(residuals(fit))
```



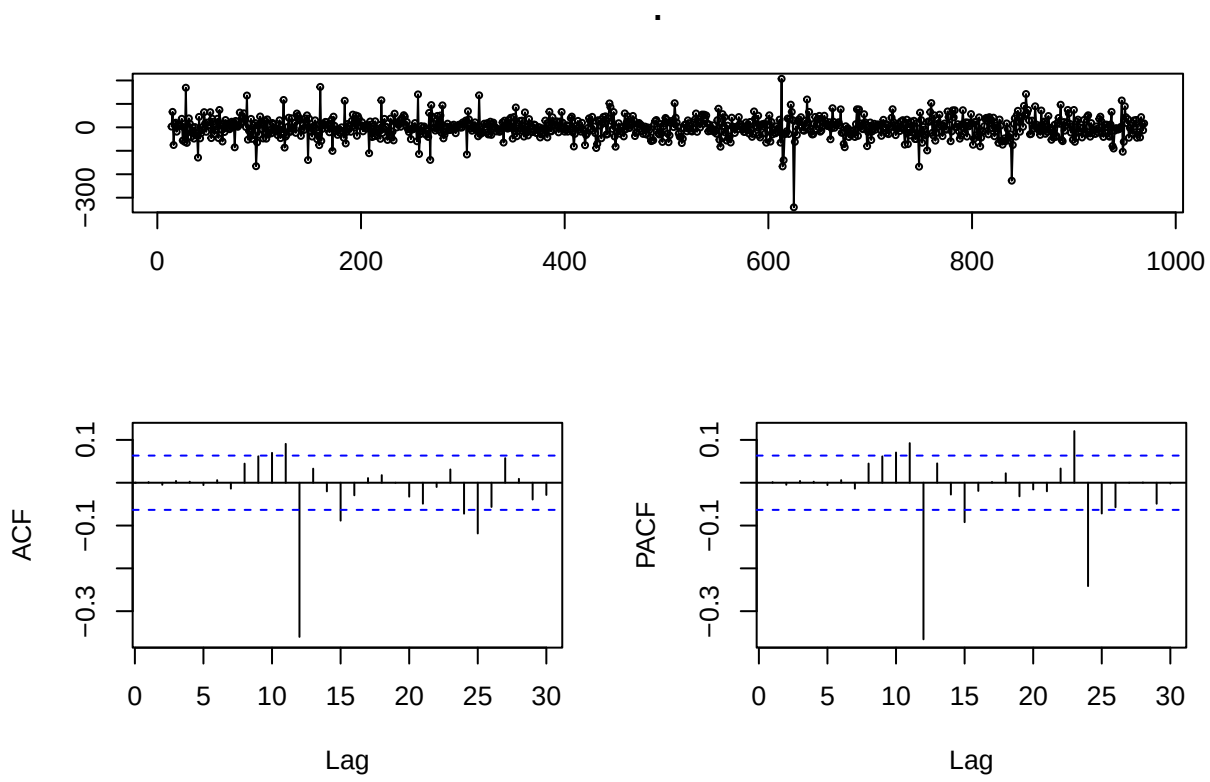

```
plot(forecast(fit,h=10),shadecols="oldstyle")
```



After taking a seasonal difference and a first difference, the auto.arima function chose an ARIMA(1,0,2) with zero mean, which is the best model according to AICc.

e)

```
residuals(fit) %>% tsdisplay()
```



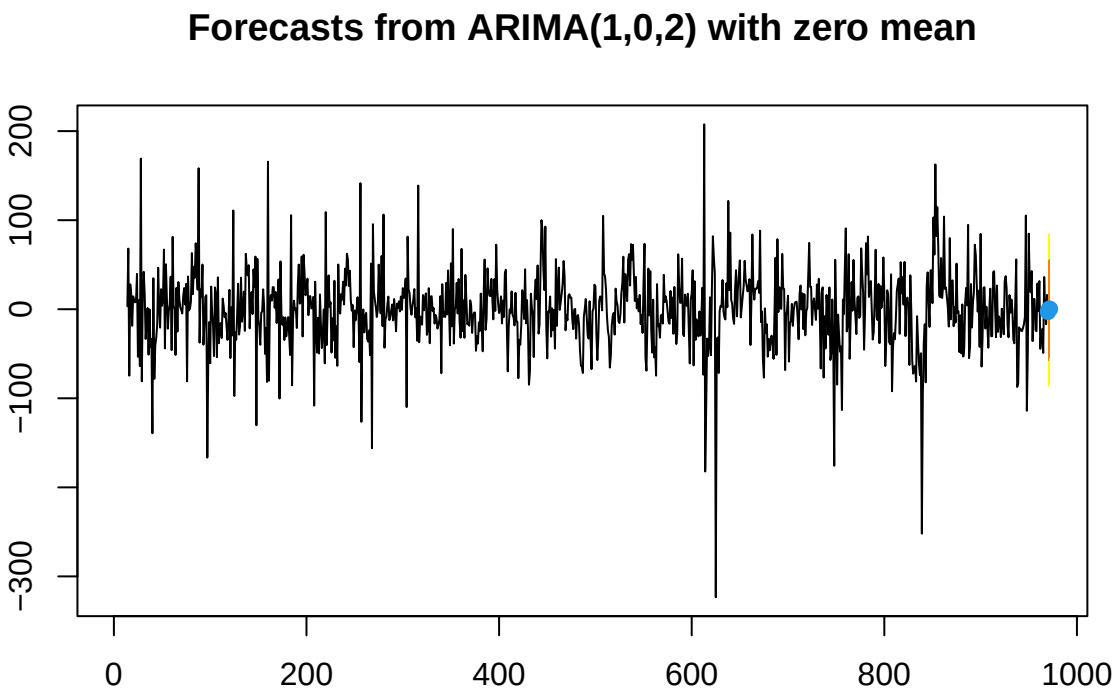

```
ljung_box(residuals(fit), lag = 12, type = "Ljung-Box")
```

```
##   lb_stat lb_pvalue
##  144.2752    0.0000
```

No, the residuals are not white noise. We can test using the Ljung-Box test. The p-value is approximately 0, so we reject the null hypothesis that the residuals are white noise.

f)

```
plot(forecast(fit,h=3),shadecols="oldstyle")
```



The forecast agrees with the data from the FRED.

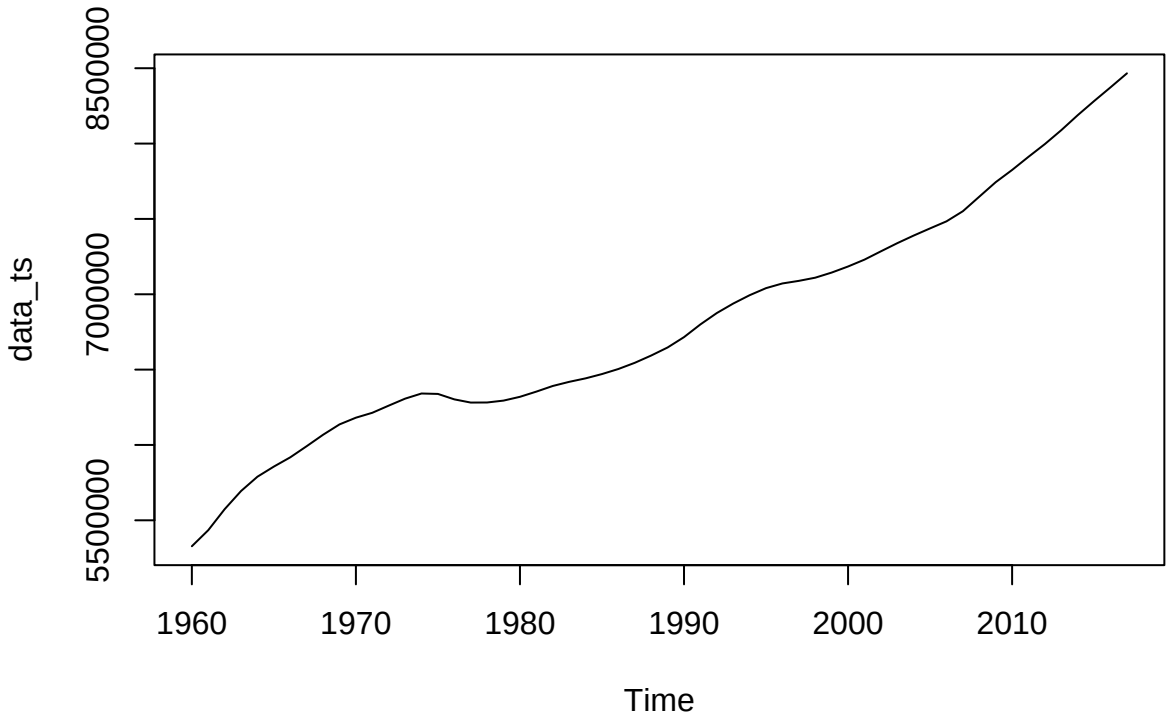
g)

After trying different years ahead, I would say staying under 10 years ahead would be ideal.

Problem 9

a)

```
data <- global_economy %>%
  filter(Country == "Switzerland")
data_ts=ts(data$Population, start= c(1960, 1), frequency = 1, end=c(2017,1))
plot(data_ts)
```

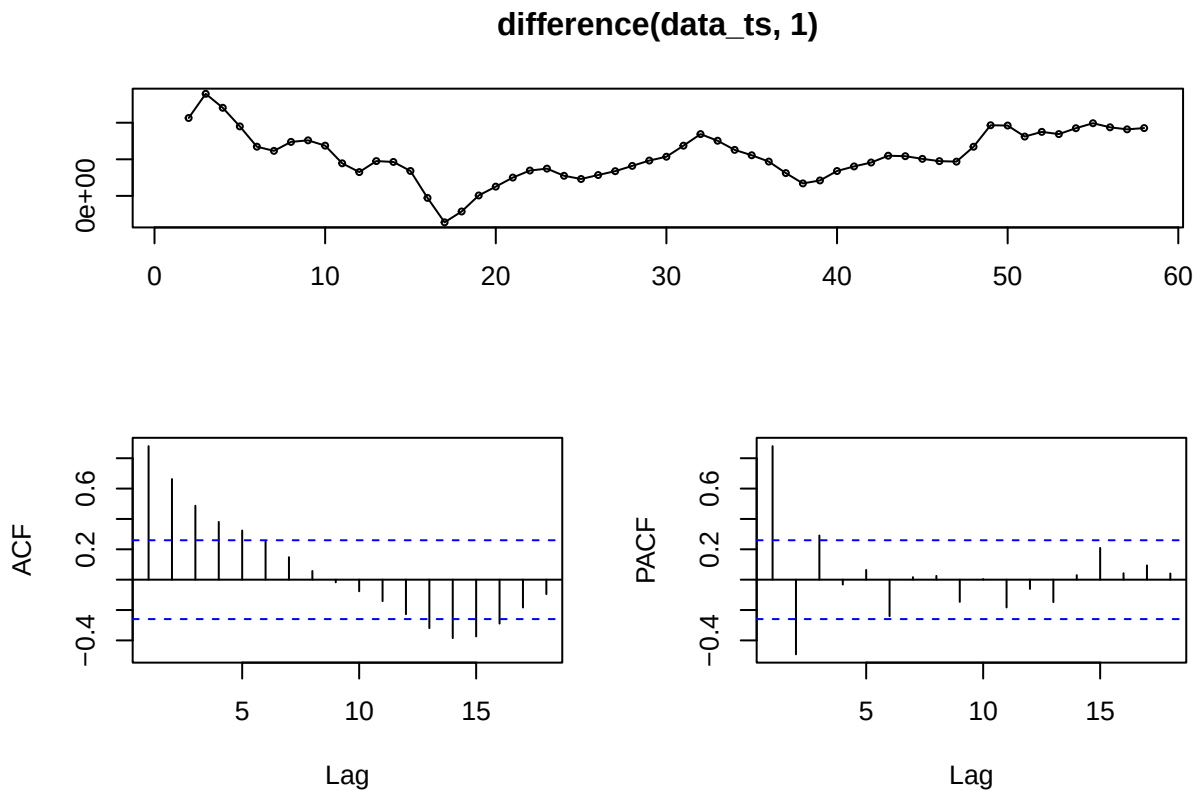


b)

This is an arima(3,1,0)

c)

```
tsdisplay(difference(data_ts, 1))
```



There is a decay in the ACF and the PACF cuts off at lag 3, indicating an AR(3).

d)

2018=8.5591 million people

2019=8.6495 million people

2020=8.7360 million people

e)

```
arma310<-arima(data_ts, order = c(3, 1, 0))
forecast(arma310,h=3)
```

```
##      Point Forecast   Lo 80   Hi 80   Lo 95   Hi 95
## 2018        8560200 8545790 8574610 8538162 8582238
## 2019        8653347 8611802 8694893 8589809 8716886
## 2020        8743772 8668294 8819249 8628339 8859205
```

```
summary(arma310)
```

```
##
## Call:
## arima(x = data_ts, order = c(3, 1, 0))
```

```
##
## Coefficients:
##          ar1      ar2     ar3
##       1.7041  -1.2352  0.5182
## s.e.  0.1270   0.2198  0.1282
##
## sigma^2 estimated as 126431317:  log likelihood = -614.95,  aic = 1237.9
##
## Training set error measures:
##                     ME     RMSE      MAE        MPE      MAPE      MASE
## Training set 973.2997 11168.74 8669.856 0.01329815 0.1319666 0.1516522
##                    ACF1
## Training set -0.068729
```

These values are slightly different because R found different parameters for the AR1, 2, and 3 components which would change the calculations.